KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

INSTITUTE OF DISTANCE LEARNING

(BSC AGRICULTURAL ENGINEERING, FOURTH YEAR)

AE 453: ENVIRONMENTAL CONTROL & LIVESTOCK HOUSING

[Credit: 2]

F. Kemausuor & A. Bart-Plange

Publisher's Information

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Course Introduction

COURSE OVERVIEW

AE 453: ENVIRONMENTAL CONTROL AND LIVESTOCK HOUSING introduces students to the general concepts on animal environment, monitoring of animal environment, control of animal environment and animal waste management.

COURSE OBJECTIVES

After going through this course, you should be able to:

1. Define and explain the general concepts of animal environment.
2. Explain modes of monitoring and controlling animal environment.
3. Identify factors that affect animal environments
4. Discuss basic concepts of animal waste management

COURSE OUTLINE

The course material has been structures into six (4) units.

- Unit 1: Heat and Moisture Production of Animals
- Unit 2: Psychrometry
- Unit 3: Environmental Control Techniques
- Unit 4: Livestock Housing

COURSE STUDY GUIDE

This provides a monthly/weekly schedule of progress of your learning.

Meeting/ Week #	Unit/Session	FFFS/Practical/Exam/Quiz
1	Unit 1/1-1 & 2-1	FFFS
2	Unit 2/1-2 & 2-2	FFFS
3	Unit 3/1-3 & 2-3	FFFS /Quiz
4	Unit 4/1-4 & 2-4	FFFS

GRADING

Continuous assessment: 30%

End of semester examination: 70%

RESOURCES

Choose an item.

You will require Click here to enter text.for this course.

To complete this course you would need to accomplish [theory], [number of laboratory/workshop/ practical/ tutorial] and [insert number of credits]

REFERENCES

1. Midwest Plan Service, 1976, Structures & Environment Handbook, Iowa State University Press, Iowa
2. David & Peter Sainsbury, 1988, Livestock Health & Housing, Alden Press, Oxford
3. J. A. Clack, 1981, Environmental Aspects of Housing for Animal Production, Butterworths Publishing
4. Clive Philips & David Piggins, 1992, Farm Animals and the Environment, CAB International, Cambridge University Press
5. Merle L. Esmay, 1978, Principles of Animal Environment, AVI Publishing Company Inc., Westport, Connecticut

READING LIST / RECOMMENDED TEXTBOOKS / WEBSITES / CD ROM

1. Curtis Stanley E. 1987, Environmental Management in Animal Agriculture, Iowa State University Press, Iowa
2. Lennart Bengtsson & James Whitaker, 1986, Farm Structures in Tropical Climates, FAO Publication, ROME, 1986

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List Of Abbreviations and Notations

Unit 1

HEAT AND MOISTURE PRODUCTION OF ANIMALS

Introduction

Unit 1 presents an overview of heat and moisture production in livestock and how this interact with their immediate surroundings.



Learning Objectives

After reading this unit you should be able to:

1. Identify the factors that affect the environment of livestock.
2. Discuss appropriate considerations for managing livestock environment.

Unit content

Session 1-1: Introductory concepts

1-1.1 Introduction
1-1.2 Environment and animal function and performance
1-1.3 Heat production & Environmental Temperature

Session 2-1: Heat and moisture production

2-1.1 Heat Production and Heat Balance
2-1.2 Animal Heat Capacity
2-1.3 Moisture production
2-1.4 Heat Transfer

SESSION 1-1: INTRODUCTORY CONCEPTS

1-1.1 Introduction

The key issues that need to be considered under animal environmental management are:

1. Adequate fresh air
2. Freedom of movement
3. Comfort of immediate environment
4. Freedom to follow innate behaviour pattern

Generally, an animal's life is controlled by the environment comprising all of the physical, chemical or biological elements or factors that surrounds the animal's body. There are various factors of the environment that may either promote or impair animal performance by facilitating or inhibiting productive and reproductive processes. The factors that influence an animal's environment include: thermal factors, light factors and factors that cause behavioural reactions. Thermal factors include: Air temperature, Relative humidity of air, Air movement (air speed), Composition of the air and Air pressure.

All farm animals are homoeothermic, exhibiting a relatively constant long-term body temperature under the most varying conditions with large variations of internal heat production due to metabolism (for maintenance and production) and variation in the surrounding thermal environment (temperature, air velocity, radiation, bedding, etc). The surrounding environment can also add heat to an animal if the ambient air temperature exceeds its body temperature or by direct or indirect solar radiation. Consequently it is essential for agricultural engineers to understand animal heat balance from the animal and physiological view. It is recognized, however, that short-term fluctuations in body temperature occur because of metabolic processes, physiological changes, and environmental stressors.

Each environmental factor varies in space and time and hence the complexity of an animal's environment. The environment of confined animals is influenced by the interplay of the following:

1. Biological activities of the animals and other operations in the enclosure.
   These include:
   - Animal heat production
   - Animal moisture production
   - Gas production (CO_2, NH_3, H_2S, CH_4 etc.)
2. Ambient (outside) environmental conditions. These are:
   - Temperature
   - Relative humidity
   - Air movement
   - Solar radiation
   - Air composition

3. Size and technical characteristics of structure. These include:
    - Dimensions, orientation and shape of structure
    - Thermal properties (heat storage and transfer abilities)
    - Porosity
    - Moisture (water vapour) diffusion properties
    - Light and sound
4. Environmental control techniques. These include technical measures put in place to control the animal`s environment:
    - Ventilation
    - Heating (e.g. in brooder houses)
    - Cooling
    - Moisture removal / addition

Consequence of a poorly designed animal house/environment

Poorly designed animal houses will have negative influence on:

- Health of farm workers and animals
- Performance and productivity
- Heat and moisture production
- Air composition (gases & odour production)
- Rate of degradation of animal house (building life)
- Income generation

1-1.2 Environment and animal function and performance

The animal is said to be an open thermodynamic system. This is because the animal continuously trades energy and matter with its environment. The external factors sometimes tend to produce changes in the internal environment of the animal.

Consistent and proper functioning of the whole animal requires steady or at least constantly equivalent conditions in the body. The animal therefore needs to maintain a steady state in its internal environment despite fluctuating external surroundings. Livestock and poultry are referred to as homoeothermic (i.e. they maintain a relatively constant body temperature). Temperature regulation is an example of Homeostasis (homeokinesis), i.e. the maintenance of a bodily state within narrow limits.

The animal is able to control its body temperature by a gland called hypothalamus gland which acts as a thermostat. The hypothalamus regulates the body temperature by detecting internal body temperature, integrating it with neural signals from thermo-receptors at the skin surface, and controlling neural and hormonal responses that alter heat production and heat losses. In this way

temperature regulation is effected. Table 1 shows typical body temperatures for some selected Homeotherms.

Table 1: Typical body temperatures for some selected Homeotherms

Homeotherm	**Body Temperature**
Man	$37^{o}C$
Cattle	$38.5^{o}C$
Horses	$38^{o}C$
Chicken	$41.7^{o}C$
Sheep	$39^{o}C$
Goats	$40^{o}C$
Swine	$39^{o}C$

A temperature regulating system analogy

A modern house with a temperature control system can be used as an analogy to the temperature regulation ability of the body. A house just as a body has an external and internal environment.

To control the temperature of a room one needs a control system, which includes:

a. Thermostat (Sensor)

b. Air conditioner (effector) or

c. Furnace (effector)

The thermostat acts as the sensor whilst the air conditioner or furnace acts as an effector. These work together to keep the internal environment almost constant regardless of its external environment.

The sensor or thermostat senses and responds to rises above and below predetermined room temperature. An effector or an electric circuit will close or open to either start the air conditioner or stop its operation. The same principle will apply to a heating system.

1-1.3 Heat production & Environmental Temperature

An animal's heat production rate & body core temperature to the zones of effective environmental temperature gives an indication of zones (Figure 1) in which the regulation of body temperature is effective.

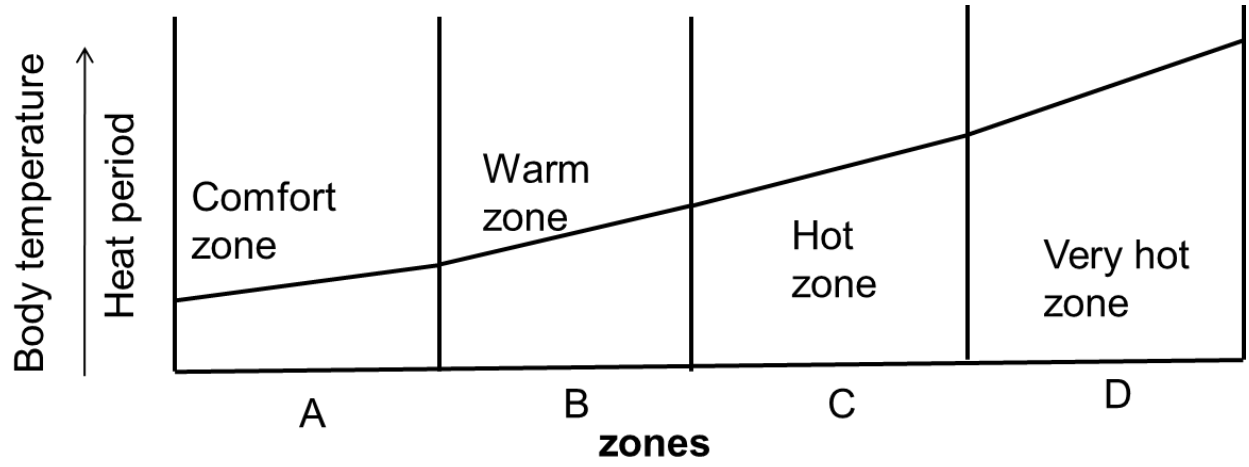


Figure 1: Relationship b/n environmental temperature, animal heat production and body temperature

Zone A: This zone is referred to as thermo-neutral or thermal comfort zone. Here:

- Effective environmental temperatures are optimal
- No heat stress is experienced in this zone
- Metabolic rate is minimal and
- Heat dissipation mechanism is not in active operation

Zone B: This is referred to as the warm zone. Here:

- Effective environmental temperatures are above the upper limit of zone A
- Heat stress starts (sweating & panting starts)
- Metabolic rate is increased
- Heat dissipation comes into play

Zone C: This is referred to as the hot zone. Here:

- Effective environmental temperatures reach upper critical limit
- Excessive heat stress is experienced (sweating & panting increases)
- Excessive metabolic rate is experienced
- Very active heat dissipation takes place
- Body temperature control is not very effective

Zone D: This is referred to as the intolerably hot zone or death zone. Here:

- Effective environmental temperatures reach unbearable limits
- Body temperature rises uncontrollably
- Death becomes imminent

Effective environmental temperature

The relationship between heat production rate and environmental temperature depends on:

- Animal's heat resistance
- Acclimatisation to heat
- Duration of heat stress and
- Degree of that heat stress

Adaptation to the environment

Strain and stress

Strain is any functional, structural or behavioural reaction to an environmental stimulus. Strain may be adaptive or non-adaptive. Stress as referred to in environmental control is any environmental situation that provokes an adaptive response. Stress may be chronic (gradual & sustained) or acute (abrupt and often profound). The relationship between stress & strain is that an environmental stress provokes animal strain. Environment stress occurs when a given animal's environment changes example as environmental temperature rises (stress factor).

Functional Priorities during Stress

Animals have three kinds of life processes:

1. Maintenance process: maintains life and ensures day to day survival of the individual animal
2. Reproductive process: Provides for perpetuation of the species.
3. Production process: Cultured to yield products such as meat, milk, eggs etc.

In some environmental situations, all life processes can be supported but in others, they cannot be, in which cases priorities must be exercised. The processes are differentially important to an animal's survival at a given moment. Hence when resources become limiting, some processes must be de-emphasized so that others can be emphasized. For example: An animal's growth rate might have to be reduced so that breathing and sweating rates can be increased in combating a hot environment. Strains are maintenance functions. When an animal strains in response to stress, its maintenance needs always increase. In practical situations, an animal`s functional capabilities often decrease at the same time.

Stress can cause:

- Sweating and panting (which are maintenance functions)

- Loss of appetite and
- Low productivity

The order of priority for the animal to maintain the life's processes under stressful condition is:

- Survival (maintenance)
- Reproduction
- Production

Strain-Stress & Productivity

Strain can influence the partition of the animal's resources among maintenance reproduction and production in five ways:

1. Strain alters internal functions
2. Strain diverts nutrients
3. Strain directly reduces production
4. Strain impairs disease resistance
5. Strain increase variability

SESSION 2-1: HEAT AND MOISTURE PRODUCTION OF ANIMALS

2-1.1 Heat Production and Heat Balance

Heat Terminology

1. Heat is a form of energy (Units: joules, calories & British thermal unit (Btu))
   1 Btu = 252 cal = 1,054 Joules of energy; 1cal = 4.187 Joules

2. Temperature: It is a measure of a body's ability to transfer heat from or receive heat from matter in contact with it or from surrounding surfaces.
   Units: Kelvin, K, Celsius or centigrade, $^{\circ}$C.

3. Ambient temp: Temperature of the medium surrounding a body or system
   E.g. room air temperature.

Inter conversion between temperature scales

°C = (5/9) [°F - 320] = [t_F - 32] /1.8

°K = °C + 273 °C

°F = (9/5) ^{o}C + 32 = 1.8t_C + 32

4. Specific heat capacity: It is the quantity of heat needed per unit mass (weight) to increase its temperature by 1°C.
   Units: J/kg°C or kCal/kg°C

5. Heat capacity: The same mass of different materials can hold different quantities of heat. Hence heat capacity is the ability of a material to absorb or hold heat. It is the number of calories which will raise the temperature of a whole body of matter by one degree. For instance water must be supplied with more heat than oil in order to produce the same rise in temperature. We therefore say that water has the greater heat capacity that oil.
   Unit: J/k or cal/°C

6. Sensible heat: It is a measure of the energy that accompanies a temperature change.
   $q_s = mc\theta = mc(t_1 - t_2)$

   Note: θ = ΔT = t_1 - t_2
   q_S - Sensible hat
   c - Specific heat capacity (J/kg°C or kcal/kg °C).
   m – Mass of substance in kilogram

7. Latent Heat Q_L: The amount of heat required or released when the material changes phase without change in temperature. The three phases are Solid, Liquid and Gas.
   The solid - liquid change involves Latent heat of fusion (L_f). The liquid - vapour change involves Latent heat of vaporisation (L_v)

   Q_l (q_l) = mL_f or mL_v,

   Units: kJ/kg, kcal/kg.

   L_f = 335 kJ/kg and L_v = 2260kJ/kg

2-1.2 Animal Heat Capacity

Animal bodies consist of water and dry matter and are therefore non-homogeneous. But if a body's ratio of water to dry matter is known, its heat capacity can be estimated. The overall heat capacity of an animal's body is the addition of its water heat capacity (C_W) and its dry matter heat capacity (C_D). Animal in the adult stage has a moisture content of about 70%.

Total Heat Content:

When a body's heat content increases, its temperature rises and vice versa. As a certain amount of heat gets into or out of a body, the body's heat content changes by that amount. This change in the heat stored by the body equals the product of the body's heat capacity and change in temperature i.e. $C_b\Delta T$

2-1.2.1 Heat Production

The major source of heat in a building that houses livestock is heat given out by livestock. This heat production is a function of body weight and metabolism. Heat produced by animals is in two parts:

(1) Sensible heat (due to ΔT)

(2) Latent heat (a measure of animal's water-vapour production)

The animal produces heat when transforming energy of food (chemicals) into work. The rate of internal heat production of an animal varies with size, body weight, breed, health, stage of growth, nature and rate of feed intake, level of production, gestation ages, degree of activity and environmental conditions.

The other sources of heat from animal houses are heaters, lights and motors, heat gain from walls, floors, ceilings etc.

2-1.1.2 Heat Losses

Heat losses from a building will constitute or consist of heat flow out of a building. Example is the ventilation heat loss which is when air is used to move heat and moisture from a building.

Standard heat production of homoeothermic animals

Heat production is related to body weight:

$$q_{st} \equiv W$$

$$q_{st} = kW^n$$

Where

q_{st} is the standard heat production in kCal/hr or J/hr etc.

W is the body weight in kg

k is metabolic rate constant and is standard heat production per metabolic body size of an animal in kCal/hr /unit MBS.

n is a constant usually given as 0.75

The animal's metabolic body size designated MBS has been found to be W^n
The average value for k is 3kCal/hr per unit MBS
Heat produced by animals will have to be transferred from the animal's body. Since heat production is a function of body weight and metabolism and heat transfer from an animal depends largely on surface area, heat transfer per unit weight is higher for smaller animals. The ratio of surface area to animal body weight decreases as weight increases. See example in Table 2.

Table 2: Surface area to weight ratios of animals

Weight	**surface area**	**surface area to weight ratio**
1.82kg	$0.18m^2$	0.176
454kg	$4.985\ m^2$	0.011

2-1.1.3 Animal total surface area (ATSA)

Animal total surface area can be estimated by 3 methods:

- Direct measurement
- Dividing the body into parts with regular geometric shapes and then determining the combined areas of the shapes
- Photometry

The total surface area of an animal has been found to be roughly proportional to two thirds of the body weight.

$A \sim W^{2/3}$ which implies that $A = k\ W^{2/3}$

Once an animal's body weight and the constant, k are known, the surface area can be estimated. Some typical equations developed by some researchers:

Brody and Coworker found for Pigs that:

$$A = 0.097\ W^{0.63}$$

Where A is the total surface area and W is the body weight

Later Kelley et al. also found for Pigs that :

$$A = 0.073\ W^{0.66}$$

2-1.1.4 Heat loss and cooling rate

An animal's heat loss rate is proportional to its surface area, which in turn is somewhat proportional to the two-thirds power of body weight. An animal's cooling rate (its rate of temperature decline) is however proportional not only directly to heat loss rate but also inversely proportional to the body's heat capacity, which in turn is directly proportional to body weight. Thus, cooling rate is proportional to negative one-third power of body weight.

Implications:

A larger animal loses heat at a higher absolute rate than a small one, but the larger animal's cooling rate would be lower.

2-1.1.5 Heat balance in an animal housing

Heat Gain/ sources

q_a - Sensible heat gain from the animals

q_l - Latent heat gain from the animals

q_m - Mechanical heat from light, motors, etc

q_{sp} - Supplemental heat from furnace, lamps, firewood etc.

Heat gain = $q_a + q_l + q_m + q_{sp}$

Where a = animals, m = motors, sp = supplementary

Heat loss = $q_b + q_v + q_e$

Where: q_b = heat loss through the building

q_v = heat loss through ventilation

q_e = heat loss through evaporation

For heat balance:

Heat gain = Heat loss

$\Rightarrow q_a + q_l + q_m + q_{sp} = q_b + q_v + q_e$

$\Rightarrow q_a = q_{bv} \pm q_v$

Note: - Ventilation heat is incorporated in the equation

- In temperate zone we consider q_{sp}

2-1.3 Moisture production

2-1.3.1 Moisture source:

The heat and moisture production at house level is based on animal level of heat and moisture production and then the influences of various factors. These factors mean that more moisture evaporates with heat from the animal's sensible heat. Factors include flooring system, stocking density, watering, moisture content of the feed and feeding system, animal activity, and relative humidity. The most significant of these factors which will be discussed here include:

1. Animal-water vapour production (calculated from latent heat production)
2. Moisture evaporation from wet building surfaces (included in animal latent heat production)

2-1.3.2 Moisture losses

Water vapour losses occur almost exclusively in the ventilation air as cool dry air replaces warm moist air. The moisture loss from the air by condensation on wall surfaces is usually small and can be neglected.

Moisture balance

$M_a W_2 = \mathrm{M_a} W_1 + M_a W_e$

Ma = ventilation rate

W_1 = incoming water vapour

W_2 = water vapour in exhaust air

W_e = water vapour from surface-water evaporation

X= Animal (pig)

Ventilation rate $q_v = m(h_2 - h_1)$

2-1.4 Heat Transfer

Basic to any discussion of insulation and ventilation is an understanding of the way heat is transferred. Heat moves from place to place by conduction, convection, radiation or some combination of these modes whenever a temperature difference exists. Heat energy always tends to transfer from high temperature to low temperature regions.

2-1.4.1 Conduction

If one end of a metal bar is placed in a fire, then although no part of the bar moves the other end will become warm. Heat energy is passed from molecule to molecule in the metal bar. Here, there is no change of molecular position. Transfer of heat energy through a material without the molecules of the material changing their position is termed conduction. Conduction can occur in solids, liquids, and gases although the speed at which it occurs will vary. The thermal conductivity is a measure of how heat is passed from particles to particles. The rate of heat flow by conduction depends on the amount of temperature difference and thermal conductivity of the material.

Conductive heat flow

Heat flow by conduction is given by

$$Q = \frac{kA(T_1 - T_2)}{d}$$

Where: d = thickness of material
A = cross sectional area
ΔT = change in temperature
K = thermal conductivity

For the animal heat flow by conduction:

$$Q_c = \frac{kA_A(T_1 - T_2)}{S}$$

Where: A_A = Animal's effective conductive surface area
K = thermal conductivity of the environmental substance
T_1/T_2 = temperature of the environmental substance at a point some distance from the animal contact point
T_2 = environmental surface temperature at point of contact in animal
S = distance between points where T_1 and T_2 are determined.

2-1.4.2 Convection

Air is a poor conductor of heat yet it is still possible to heat all the air in a room from a heat panel or from a heater. This is by the process of convection. A heated liquid or gas (fluid) actually moves from one place to another carrying its heat with it. We therefore say that by definition convection is the transfer of heat energy through a material medium by bodily movement of particles. The rate of heat flow depends on the temperature of the moving fluid and rate of flow.

For animals we have

$$q_c = hA_A(T_a - Ts)$$

Where: A_A = Animal's effective convective surface area

T_a = air temperature

T_s = surface temperature of animal

S = distance between points where T_1 and T_2 are determined.

h = convective coefficient

NB: h depends on air velocity and animal surface area.

2-1.4.3 Radiation

Heat is transferred from the sun to the earth through space where conduction and convection is not possible. The process of radiation is responsible for this heat transfer through space. Radiation is the transfer of heat energy by electromagnetic waves. Heat radiation occurs when the thermal energy of surface atoms in a material generates electromagnetic wave emanates from a hot body and can travel freely only through completely transparent media.

Heat cannot move by radiation through opaque material, but instead is partially absorbed and reflected from their surface. The atmosphere, glass and translucent material allow a substantial amount of radiant energy to pass through while absorbing and reflecting some. Note that:

- Dull black surfaces are best absorbers and emitters.
- Shining surfaces are poor absorbers and emitters.
- For animals we have, heat flow by radiation

Heat flow by radiation in animals is given by

$$q_r = A_r a \sigma T_e^{\,4} - A_r e \sigma T_s^{\,4}$$

Where:

A_r = effective radiant surface area

a = absorbtivity construction of animal surface

e = emmissivity construction of environmental surface

σ = Stefan Boltzman constant

Te = absolute average temperature of animal radiant environment

Ts = absolute average temperature of the animal radiant surface

2-1.4.4 Insulation and heat flow

Insulation is used to reduce the flow of heat into or out of a structure. Although all building materials have some insulating values, the term insulation material is generally reserved for a family of product designed to provide this service. The type of wall construction and type and amount of

insulation material used will determine what the heat flow rate will be for a particular wall or building. A knowledge or relative insulation value of different construction methods and insulation materials and how to put them together to estimate the overall insulating material value of a wall would be a great help for selecting appropriate insulating material for a particular application.

Insulators are generally classified as follows:

1. Rigid pre-formed materials. (e.g. aerated concrete block)
2. Loose fill materials (e.g. cork granules in ceiling)
3. Reflective materials (e.g. aluminium foil)
4. Material formed on site (e.g. plastic foam)
5. Flexible materials

Thermal resistance

Building material (B.M) have been characterised according to their insulating ability. The insulating ability of a material or the resistance of a material to the flow of heat through it is called the thermal resistance.

Materials

Homogeneous material: materials that are the same throughout (e.g. wood and concrete). The thermal resistance (R) of homogeneous material is expressed per unit of material thickness.

Non homogeneous materials

Materials that are not the same throughout (e.g. concrete block) and the thermal resistance is expressed for a particular thickness. Thermal resistance values have units of m^2k/W or mk/W

Total thermal resistance of an element

When the building element is made up of two or more different materials the thermal resistance can be estimated with the knowledge of the individual resistance.

$R_T = R_{S1} + R_1 + R_2 + \ldots + R_{S0}$

Where R_T = Total thermal resistance of composite element

R_1, R_2 = Resistance for homogeneous material

R_{S1}, R_{S0} = Resistance of inside and outside surfaces

Overall heat Transfer Coefficient (U-value)

It is the heat flow in watts through 1 m^2 of construction where a temperature difference of 1 $^{\circ}C$ exists between outside and inside temperatures.

The U-value which is also known as thermal transmittance summarises into a simple number the somewhat complex heat transfer mechanism that occur through a construction. The U-value is related to R_T: $U = \frac{1}{R_T}$

R_T = Total thermal heat resistance of the wall ($m^2\,^oC/W$)

U = Overall coefficient of heat transfer for the wall ($W/m^2\,^oC$)

Heat Transfer in a Building

1. The general procedure for determining the total heat flow for a building or a room is to find from tables or calculate the total thermal resistance of each type of surface of the building.
2. Find a specified area of each type of surface
3. For a specified inside and outside temperatures calculate the heat flow through each surface using the heat flow formula, Q = UAT
4. And finally add the heat flow for all the surfaces to obtain the total heat flow. This procedure can best be accomplished in tabular form to simplify the area determination and to prevent overlooking any surface.

Steps:

1. Building dimensions - length, width, wall height
2. Area - ceiling, wall, floor, door and windows area
3. R or U-values: for ceiling, wall, floor, door, window
4. Building heat loss: for ceiling, wall, floor, door, window and find the sum

Temperature gradient through wall surfaces

Basically, the temperature at any point in a structure can be determined with the following expressions:

$T_X = T_i - R_X/R_T\,(T_i - T_o)$

Where T_X = temperature at any point in the wall
T_i = inside room air temperature
T_o = outside air temperature
R_X = total thermal resistance from inside to the point x
R_T = total thermal resistance from inside to the outside of the wall

The temperature drop is given by $R_X/R_T(T_i - T_o)$



Self Assessment 1-1

1) Find the heat capacity of a 159kg cow, given that the specific heat capacity of water ($C_w = 1KCal/Kg$ ℃ and that of the dry matter $C_D = 0.4KCal/Kg$ ℃). Moisture content of the animal is 75%.
2) If the temperature of an animal with heat capacity $150KCal/$ ℃ drops 2℃, the body will loose some amount of heat. If the specific heat capacity of water,$C_W = 1KCal/Kg$ ℃) and that of the dry matter, $C_D = 0.4KCal/Kg$ ℃), what is the energy loss?
3) The exposed wall of a single sheet steel tank has an area of 4 m^2 and thickness of 10mm. the temperature of the liquid is 100℃ and the air temperature is 20℃. Determine the rate of heat flow through the tank, if the thermal conductivity of the steel is 40W/m℃.



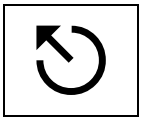
Answer tips

1) 127.5 kCal/^{o}C
2) 40 kCal
3) 1280 kW



Learning Track Activities

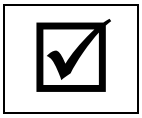
Take a learning tour of livestock houses in your community and observe the interplay of the issues we have discussed in this unit.



Unit Summary

In this unit, we have introduced environmental concepts in livestock housing. We have discussed the activities that result in moisture and heat production in the immediate environment of livestock and how these affect the general growth performance of

livestock. We have also discussed extensively the various terminologies in analysing heat production in livestock houses.



Key terms/ New Words in Unit

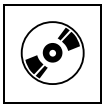
1. Homeostasis: the maintenance of a bodily state within narrow limits.
2. hypothalamus gland: a gland which helps control animal body temperature
3. Homoeotherms: Species that exhibit a relatively constant long-term body temperature under the most varying conditions with large variations of internal heat production due to metabolism (for maintenance and production) and variation in the surrounding thermal environment (temperature, air velocity, radiation, bedding, etc).
4. Heat capacity: The same mass of different materials can hold different quantities of heat.
5. Sensible heat: It is a measure of the energy that accompanies a temperature change.
6. Latent Heat: The amount of heat required or released when the material changes phase without change in temperature.





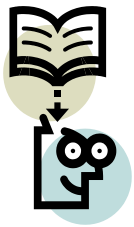






Discussion Question:

Having read unit 1, discuss recommendations for the improvement of environmental conditions in poultry houses in Ghana.



Unit Assignments 1

1. Determine the total heat flow for a structure, 2.6m by 20m long by 10m wide. The building has two wooden doors 2.5cm thick & 1.5m x 2.2m, and 10 windows (single panel), 0.75m x 1m. The wall is made of 20cm light weight concrete block with core filled. The ceiling is 1cm plywood with 15cm insulation between the beams. Temperature at the top of ceiling is the same as that of the outside wall.

   Consider the inside and outside temperatures to be 32 and 25^{0}C respectively.
   U-values:
   Wall: 2.9 W/m^2 ^{0}C
   Door: 2.9 W/m^2 ^{0}C
   Window: 6.0 W/m^2 ^{0}C
   Ceiling: 1.47 W/m^2 ^{0}C (Neglect the insulation)

2. The walls of a pig house are constructed from a single 100mm solid block of thermal resistance 0.14m2°C/W with 50mm polystyrene insulation of thermal resistance 1.25m2C/W. find the overall heat coefficient.

3. An animal housing structure has dimension of 2.6m high, 20m long and 10m wide. The building has two wooden doors 1.5m×2.2m, 0.75m×1m. the wall is made of concrete block and the ceiling is made of plywood. The temperature difference between the inside and outside is 5°C.

   a) What is the total heat transfer through the walls?
   b) What is the total heat transfer in the building?

   Take U- Values for the various components as follows:
   Wall – 3.5W/m^2°C
   Window – 6.5W/m^2°C
   Door – 3.2W/m^2°C
   Ceiling – 2.05W/m^2°C

4. A farm has 60 pigs housed in a 10m×40m×3m structure with 20m^2 of windows and 12m^2 of doors. It is 5°C warmer inside and the heat produced per pig is 600W. U-values in W/m^2°C are: window-5.8, door-1.5, ceiling-0.5 and wall-0.75. What percentage of the heat production should be removed by ventilation for heat balance to occur?

Unit 2

PSYCHROMETRY

Introduction

Psychrometry is important in the discussion of gas-vapour mixtures which are important for the design of ventilation systems and drying of agricultural produce. In this Unit, we discuss the application of the psychrometric chart in areas such as sensible heating and cooling, evaporative cooling, heating & humidification and cooling & dehumidification.



Learning Objectives

After reading this unit you should be able to:

1. Define the various properties of atmospheric air
2. Explain the application of the psychrometric chart
3. Discuss the applications of evaporative cooling

Unit content

Session 1-2: Introduction to psychrometry

- 1-2.1 Definition of physical and thermal properties of atmospheric air
- 1-2.2 The Psychrometric chart

Session 2-2: Application of Phychrometric Chart

- 2-2.1 Sensible heating
- 2-2.2 Sensible cooling
- 2-2.3 Evaporative cooling
- 2-2.4 Heating and Humidification,
- 2.2.5 Cooling and Dehumidification
- 2-2.6 Sweating / Panting Relationship
- 2-2.7 Tonne of refrigeration

SESSION 1-2: INTRODUCTION TO PSYCHROMETRY

Psychrometrics or psychrometry are terms used to describe the field of engineering concerned with the determination of physical and thermodynamic properties of gas-vapor mixtures. Psychrometry often referred to as hygrometry is the study of mixtures of air and water vapour. The atmospheric air has physical and thermal properties. These properties are used to specify the amount of water vapour in the air. They are also related to one another. These relationships between different types of measurements can be expressed in the form of table of values or the form of graphs. A psychometric chart (Figure 2) is a set of graphs which are combined so that they plot the relationships between the different variables used to specify humidity. The chart graphically expresses how various properties relate to each other, and is thus a graphical equation of state. Charts for agricultural applications are usually corrected to standard atmospheric pressure of 101.325 kPa. However charts for other elevations are available.

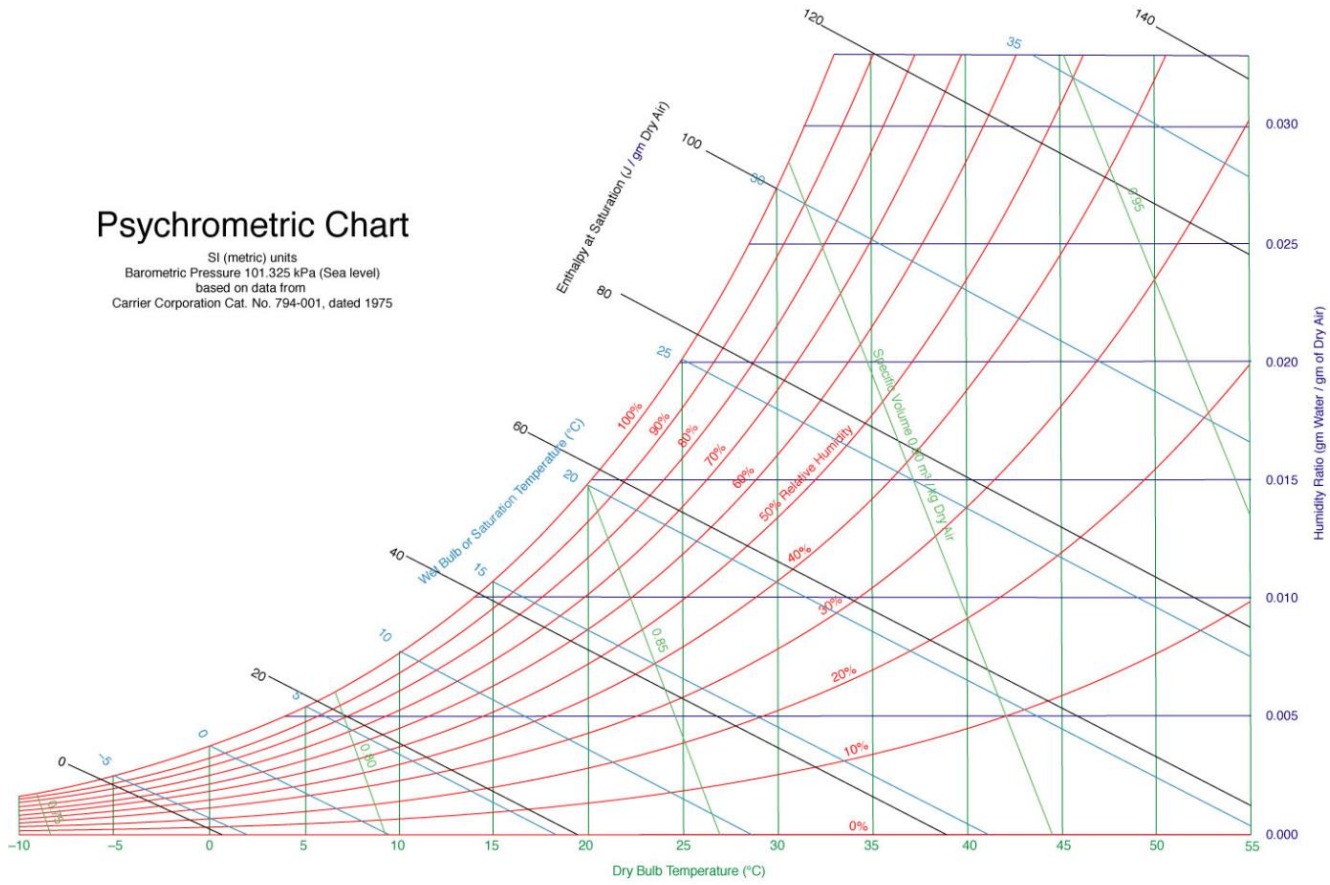


Figure 2: Psychrometric Chart

Information obtained from a psychrometric chart concerning the properties of air is used in;

1. The design of ventilation systems for buildings
2. Air conditioning systems
3. Prediction of condensation on buildings surfaces
4. Grain drying

1-2.1 Definition of physical and thermal properties of atmospheric air

The properties of air which are normally found on a psychometric chart are defined and described as follows (Please note the symbols for subsequent application):

1. **Dry Bulb Temperature (T_{db})**

The dry bulb temperature is the temperature of air measured with a standard thermometer with minimum relative motion between the dry bulb thermometer and the surrounding air. The dry bulb temperature is expressed in degree Celsius (^oC).

2. **Wet bulb Temperature (T_{wb})**

The wet bulb temperature occurs when the bulb of a standard thermometer is rapped in a wetted material (wick, gauze). The evaporation of moisture around the gauze absorbs latent heat and will cool the bulb. The wet bulb will then record a lower temperature than an adjacent dry bulb thermometer. This lower temperature is referred to as the wet bulb temperature. The wet bulb temperature is always lower than the dry bulb temperature except when the relative humidity of the air is 100%. The wet bulb temperature is expressed in degree Celsius (^oC).

3. Dew Point Temperature (T_{dp})

i. It is the temperature at which moisture (water vapour in the air) will begin to condense out of the air when the air is cooled.
ii. It is also the temperature at which the water vapour pressure is equal to the saturated vapour pressure of that air. If the temperature is reduced below this value condensation will begin to occur.

4. Relative humidity (RH)

i. It is the relative capacity of air at a particular temperature to hold moisture. R.H. is expressed as a percentage with values ranging from 0-100%. At 100% relative humidity the air is considered saturated and cannot hold more moisture.
ii. R.H. can also be expressed as the ratio of the actual vapour pressure to the saturated vapour pressure at the same temperature.

$$RH = \frac{A.V.P}{S.V.P} \times 100\%$$

5. Moisture Content (Mixing/humidity ratio (w))

i. It is the amount of water that is held in the air in the form of water vapour. Moisture content is expressed as (kg water/kg dry air).

ii. It is also the ratio of mass of water vapour to the mass of dry air in which the water vapour is associated.

6. Enthalpy (total heat content (h))

The heat energy of an air water vapour mixture. It is the total heat energy in the air which includes the heat due the temperature of the air (indicated by dry bulb temp.) and the heat energy in the water vapour itself which is the latent heat of vaporisation.

7. Specific Volume (*v*)

It is the volume of space occupied by a kilogramme of dry air at standard atmospheric pressure expressed in m^3/kg dry air.

1-2.2 The Psychometric Chart

Any particular air condition is represented by a single point on the psychometric chart. This point is known as the state point. Thus the first step at determining the properties of air will be to locate a state point for that particular air condition. Once that point has been located, one will then have to move along or move parallel to one or another of the physical property lines passing through the state point to the appropriate scales. The values are then read. It is important to note that to be able to locate a state point one needs just two properties of the air. Climate classification on the psychrometric chart is shown in Figure 3.

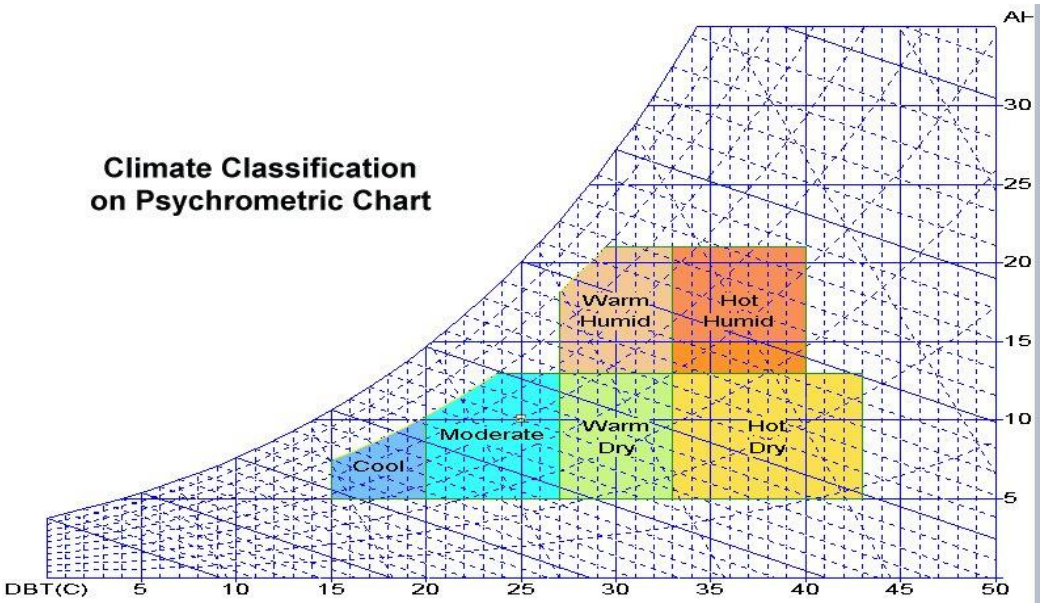


Figure 3: Climate classification on the psychrometric chart

SESSION 2-2: APPLICATION OF PSYCHROMETRIC CHART

2-2.1 Sensible heating (no change in humidity ratio)

This is the adding heat to air without changing its humidity ratio. e.g. heating for grain drying, heating of room air in livestock and poultry houses and in especially brooder houses. The following changes occur in this process:

1. Decrease in relative humidity
2. Increase in dry bulb, wet bulb temperatures
3. Increase in specific volume
4. Increase in enthalpy

The following however remain constant:

1. Humidity ratio
2. Dew point
3. Vapour pressure

This process takes place in a horizontal line and moves from left to right on the psychometric chart as shown in Figure 4.

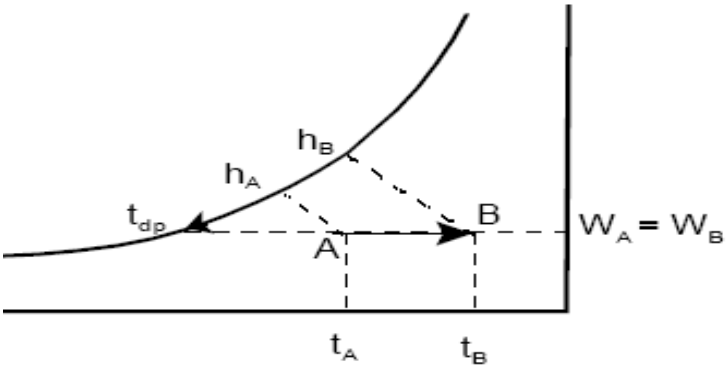


Figure 4: Sensible heating

The sensible heat involved in this process can be calculated from the mass and enthalpy change of the air.

$$q_{12} = m_a(h_B - h_A)$$

where: m_a = mass of air (dry), kg (dry air)
h_A, h_B = enthalpies of dry air (KJ/kg air)

2-2.2 Sensible cooling (no change in humidity ratio)

Sensible cooling (Figure 5) is cooling at constant humidity ratio. Example is air passing over a cooling coil with surface temperature about the dew point temperature of the air.

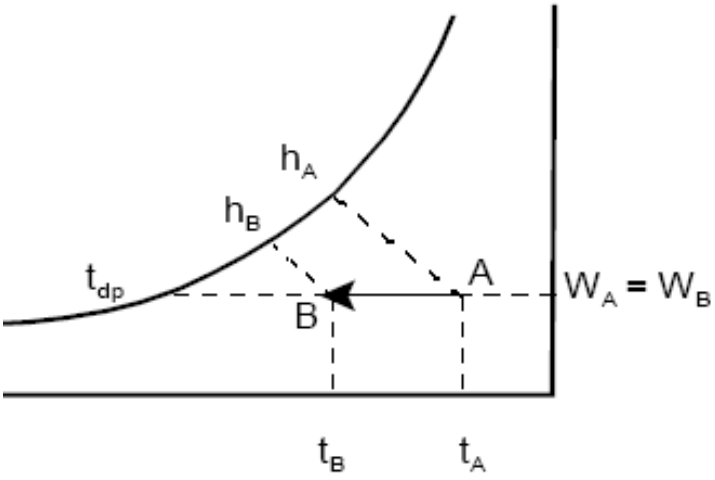


Figure 5: Sensible Cooling

Note: The final temperature cannot be below the initial dew point temperature. When that happens the process is no more sensible cooling.

2-2.3 Evaporative Cooling

Evaporative cooling is an adiabatic saturation process (no heat gain or loss) and follows upwards nearly along a constant wet bulb temperature line. Air to be cooled is brought in contact with water at a temperature equal to the wet bulb temperature of the air. The sensible heat of the initial air evaporates the water lowering the air's dry air temperature. Sensible heat is converted to latent heat and the process is an adiabatic one. Evaporative cooling (Figures 6 and 7) is most effective in hot dry climate where wet bulb depression is large.

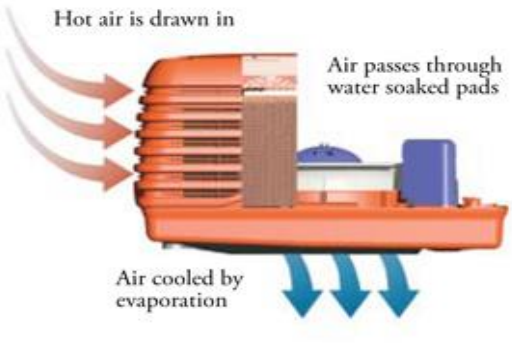


Figure 6: Combining a natural process - water evaporation - with a simple, reliable air-moving system

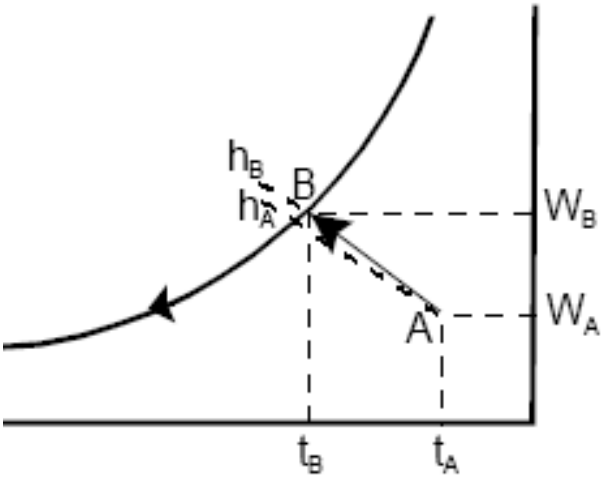


Figure 7: Evaporative cooling

2-2.4 Heating and Humidification

Heating and humidification (Figure 8) of ventilation air occurs as it moves through livestock buildings. Livestock and poultry produce heat and water vapour. Both sensible heat and water vapour are added to ventilation air. The heating and humidification of the air is best considered by looking at the two processes sequentially. The first, from state 1 to state 2, is the sensible heating that occurs when the air passes through the heat exchanger. The second, from state 2 to state 3, is the humidification process.

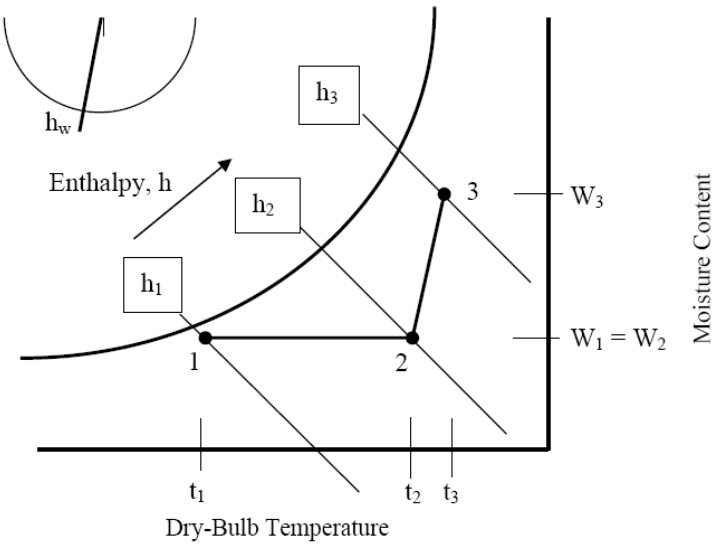


Figure 8: Heating and humidification

2-2.5 Cooling and Dehumidification

In air-conditioning, air passes over a cold, finned-type evaporator cooling coil of a refrigerator unit. The air is cooled below the dew point T^o and moisture condenses. Unless reheating or saturation occurs the final RH of the moist air is always higher than at the start. Both sensible and latent heat are removed from the air in this process. The process is shown in Figure 9. Figure 10 shows a summary diagram of all the process discussed in the session as they appear on a psychrometric chart.

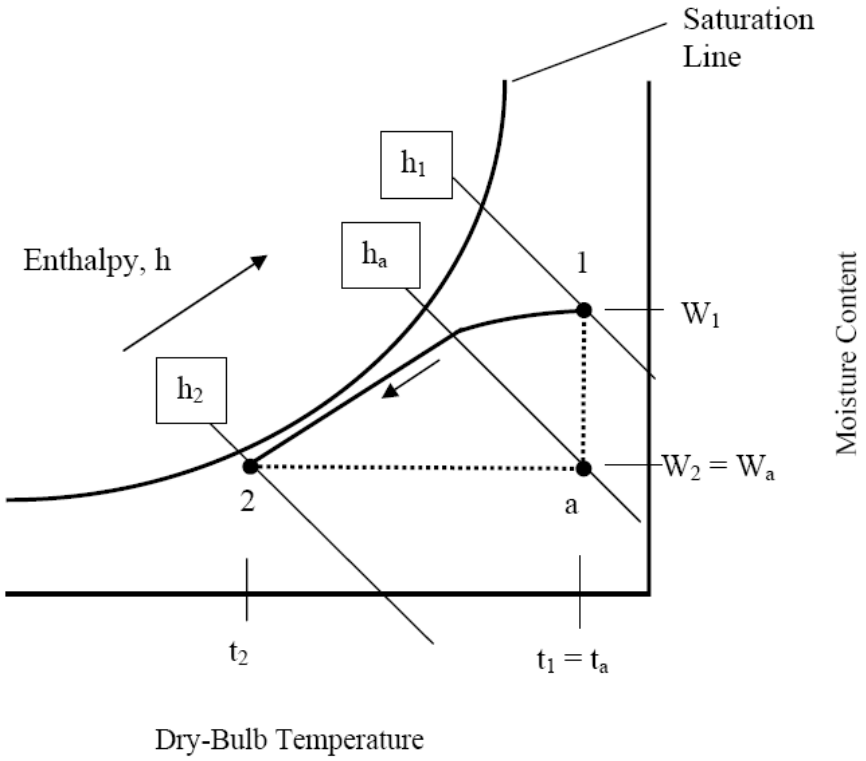


Figure 9: Cooling and dehumidification

Sensible cooling $q_s = (h_0 - h_2)$

Latent cooling $q_l = (h_1 - h_0)$

$q_{total} = q_s + q_l = (h_1 - h_2)$

Heat removed = Cooling required

= h_1 - h_2

= refrigeration per kg.

Total refrigeration $= m_a(h_1 - h_2)$

= mass flow rate x refrigeration per kg.

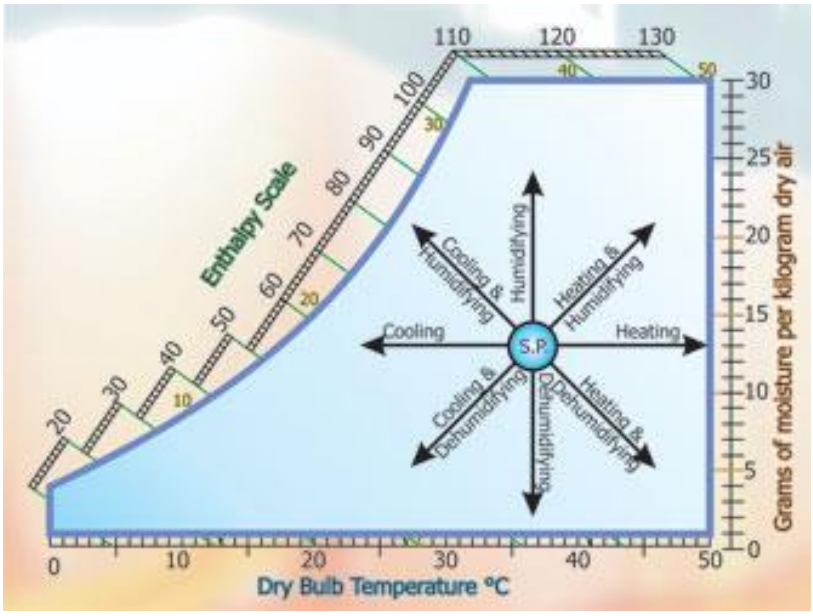


Figure 10: Summary of Psychrometric Chart Applications

2-2.6 Sweating / Panting Relationship

Temperature - humidity Index (THi) is used to indicate how comfortable the environment is for all classes of animals: human, cattle, sheep, pigs, poultry

For cattle; $THi = (.035(t_{db}) + .065(t_{wb}))$

NB

Once wet bulb and dry bulb temperatures are known this index can be obtained which can be used to tell how comfortable the animal is.

If THi < 23.5, there is no strain or little strain

THi = 25, there is moderate/removable strain.

THi > 26, there is severe strain.

2-2.7 Tonne of Refrigeration

This is the amount of heat absorbed by 1 ton of ice. One ton of refrigeration is given by 210KJ/min



Self Assessment 1-2

1. When moist air cools below its Twb. Condensation occurs. True or False?
2. Condensation occurs only below the dew point temperature. True or False?
3. In sensible cooling, the final temperature cannot be below the initial dew point temperature. True or False?
4. RH increases with evaporative cooling. True or False?
5. If RH is high and the temperature is low, Chilling occurs. True or False?

For question 6 to 8

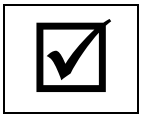
The air temperature at noon is found to be 24℃ and the dew is 12℃. Later in the day the temperature drops to 20℃ without any change in humidity ratio. Find the:

6. Relative humidity of the final state.
7. Initial wet-bulb temperature.
8. If the initial air above have been cooled evaporatively to saturation what would have been the final temperature



Answer tips

1. Read text
2. Read text
3. Read text
4. Read text
5. Read text
6. 60%
7. 12 ℃
8. 16.5℃



Key terms/ New Words in Unit

1. Psychrometry
2. Sensible Heating

3. Sensible Cooling
4. Evaporative Cooling
5. Humidification
6. Dehumidification
7. Dry Bulb Temperature
8. Wet Bulb Temperature
9. Wet Bulb Depression



Review Question:

1. Explain briefly the following properties of air found on a psychometric chart given their respective units
   i. humidity ratio
   ii. relative humidity
   iii. specific volume
   iv. wet-bulb temperature

2. The air temperature at noon is found to be 24 $^{\circ}C$ and the dew is 12 $^{\circ}C$. Later in the day the temperature drops to 20 $^{\circ}C$ without any change in humidity ratio. Find the:
   a. Relative humidity of the final state. [60%]
   b. Initial wet-bulb temperature. [12.0°C]
   c. If the initial air above have been cooled evaporatively to saturation what would have been the final temperature [16.5°C]



Unit Assignments 2

1) Moist air in an animal house at 30°C (t_{db}) and 70% relative humidity is cooled to 20°C(t_{db}).
   i. Determine the dew point temperature
   ii. Would you expect condensation to occur?

2) Air initially at 30 °C dry- bulb and 60% relative humidity is cooled to 20 °C.

   Determine:

   i. the final relative humidity
   ii. dew point temperature
   iii. the final enthalpy
   iv. Would you expect condensation to occur?

3) Determine the amount of heat removed from 150m3 of air initially at 35db and 20wb which is cooled to 25°C dry bulb temperature without changing the moisture content.

4) How many tonnes of refrigeration are required to cool 250m3 of air at 35oC db and 24oC dew point to 20oC db and 100% RH. How much moisture is removed? What is the ratio of latent to sensible heat removed in this process? A tonne of refrigeration has a cooling capacity of 210kJ/min

Unit 3

ENVIRONMENTAL CONTROL TECHNIQUES

Introduction

This unit discusses the techniques used in controlling the environment of livestock in their housing structures.



Learning Objectives

After reading this unit you should be able to:

1. Identify environmental cooling techniques.
2. Discuss evaporative cooling systems
3. Discuss ventilation and the different types

Unit content

Session 1-3: Environmental control techniques

1-3.1 Cooling
1-3.2 Heating

Session 2-3 Ventilation

2-3.1 Natural ventilation
2-3.2 Mechanical ventilation

SESSION 1-3: ENVIRONMENTAL CONTROL TECHNIQUES

Environmental control techniques that can be employed to modify the environment to make it suitable for livestock occupation include:

1. Cooling
2. Heating
3. Moisture addition or removal

4. Ventilation

1-3.1 Cooling

Cooling can be in two ways

1. Mechanical air conditioning
2. Evaporative cooling

Evaporative cooling is an affordable alternative to mechanical air-conditioning. It is an adiabatic saturation process and can work effectively where the relative humidity is low. Example is in a dessert area, airy area, arid area etc.

There are two kinds of **evaporative systems** available:

1. **Fan and pad system** (exhaust system)

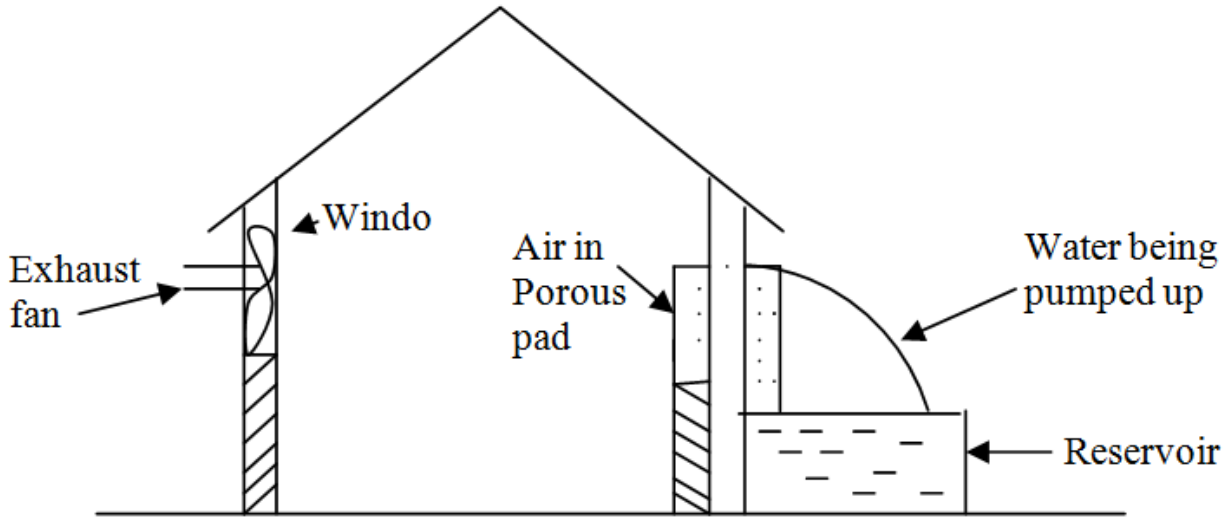


Figure 11: Fan and pad evaporative cooling system

Incoming air enters the house through the porous pad provided which has water tripling through it at all times that it is operating as shown in Figure 11. Air going through the wet pad is cooled as a result of evaporation. This air is also humidified making conditions more comfortable. The size of evaporative pad needed depends on the kind of pad material used and the ventilation rate required.

The following should be noted:

- When R.H outside increases the effectiveness of the evaporative cooling system decreases
- The dry bulb temperature of air leaving an evaporative cooler is a function of the outside dry bulb temperature and R.H

2. **Ducted evaporative cooling system** (Pressure system)

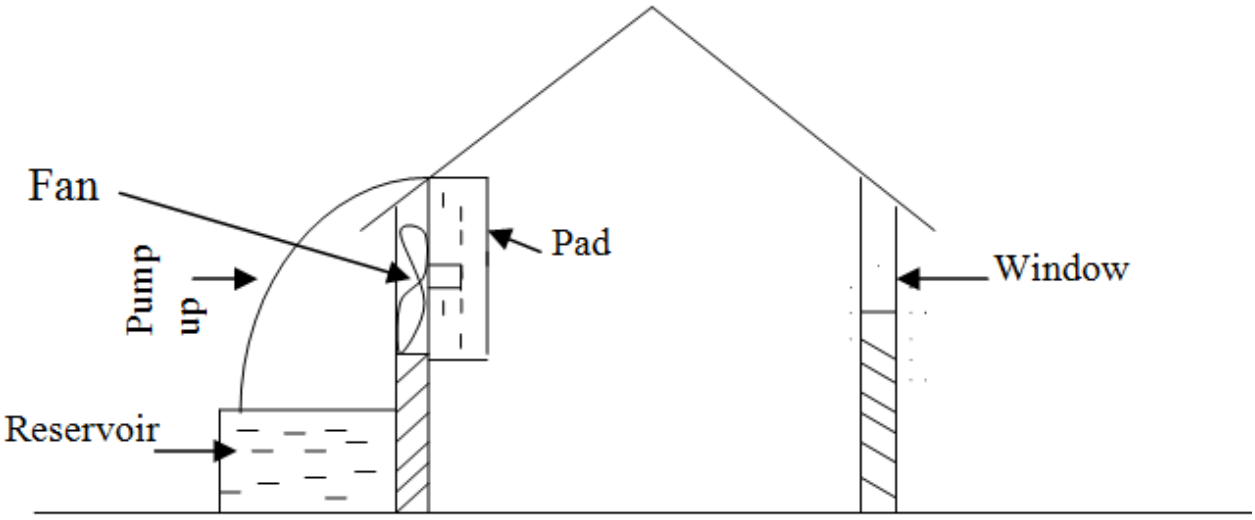


Figure 12: Ducted Evaporative Cooling System

In this system, a fan sucks outside air and blows it through a wet pad into the room (Figure 12). In the process, evaporation occurs and the room is made cooler. It should be noted that the pad must be kept wet at all times; this can be done by means of water from a water pump.

Mechanical Air Conditioning

Air conditioning is normally used to control temperature and humility for homes, offices, shops etc. In any air conditioning systems the following component may the needed:

1. Refrigeration equipment (unit)
2. Fan
3. Filters
4. Grille
5. Control

There is the need to have air condition of air i.e. controls. They are to control the correct degree of evaporation from the skin.

How air conditioner works

Air from outside and returned air combines and goes through the filter provided. It is then cooled as it goes over the (evaporative) cooling coil. By reducing the temperature to below the dew point, water condenses. The cooled air is blown through the openings into the room. This cycle is repeated. See Figure 13.

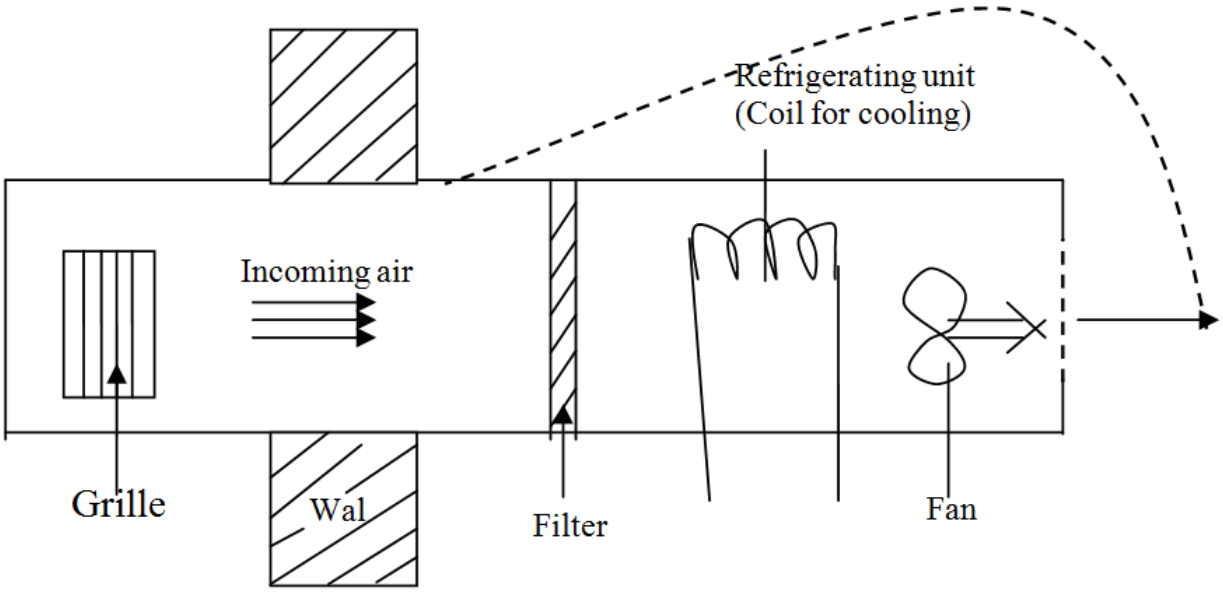


Figure 13: Typical air condition system

1-3.2 Heating

Heating can be done in the following ways:

Central Heating System

A central heating system provides warmth to the whole interior of a building or portion of a building from one point to multiple rooms. Central heating differs from local heating in that the heat generation occurs in one place, such as a furnace room in a house or a mechanical room in a large building.

Radiator

Radiators (Figure 14) are heat exchangers used to transfer thermal energy from one medium to another for the purpose of cooling and heating. Radiators are commonly used to heat buildings. In a central heating system, hot water or sometimes steam is generated in a central boiler, and circulated by pumps through radiators within the building. There are two types: Single-pipe and double-pipe. The single-pipe radiators work with steam, while the double-pipe radiators work with steam or hot water.



Figure 14: Radiator

Solar Energy

Solar energy (light and heat) has been harnessed by humans using a range of ever-evolving technologies (See Figure 15). Solar technologies are broadly characterized as either passive solar or active solar depending on the way they capture, convert and distribute solar energy. Active solar techniques include the use of photovoltaic panels and solar thermal collectors to harness the energy. Passive solar techniques include orienting a building to the Sun, selecting materials with favourable thermal mass or light dispersing properties, and designing spaces that naturally circulate air.

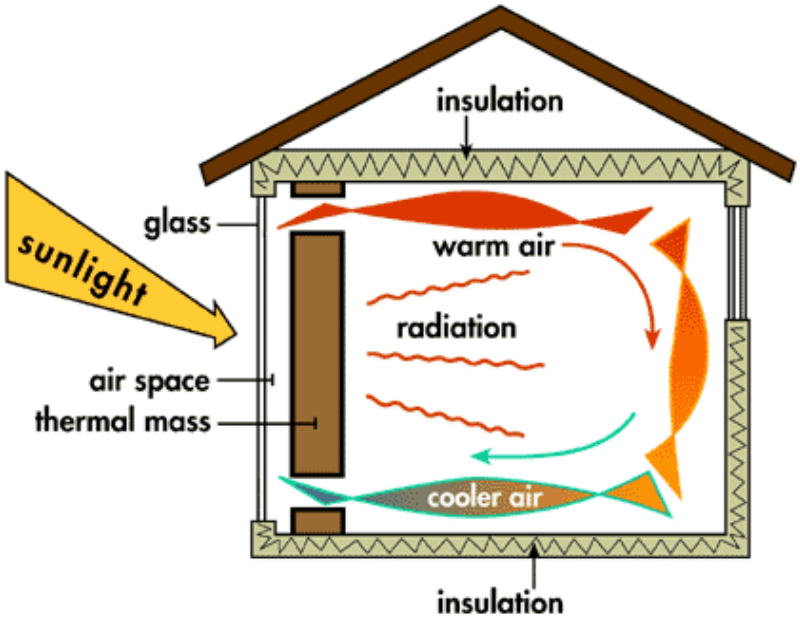


Figure 15: Solar Heating System

Boiler

A boiler is a closed vessel in which water or other fluid is heated. The heated or vaporized fluid exits the boiler for use in various processes or heating applications. The source of heat for a boiler is combustion of any of several fuels, such as wood, coal, oil, or natural gas. Most boilers produce steam to be used at saturation temperature; that is, saturated steam

.



Self Assessment 1-3

1. Discuss the exhaust evaporative cooling system.
2. Describe with the aid of a diagram how an air conditioner works.



Answer tips

1. Read session 1-3.2
2. Read session 1-3.2

SESSION 2-3: VENTILATION

Ventilation is the air exchange process that brings fresh air into a livestock building, thoroughly mixes the air, and exhausts moist contaminated air from the building. Fresh air is required in buildings to alleviate odours, to provide oxygen for respiration, and to increase thermal comfort. A ventilation system is more than just a fan. There are two types of ventilation: natural ventilation and mechanical or forced ventilation.

There are two main types of ventilation

1. Natural ventilation
2. Mechanical or force ventilation

2-3.1 Natural Ventilation

Natural ventilation is the process of supplying and removing air through an indoor space by natural means. All ventilation systems once were natural but, with the availability of electricity on farms and a blossoming control technology, mechanical or fan-powered ventilation have been

introduced. This shift from natural to mechanical ventilation took place mainly because building designers felt that the performance of mechanical systems was predictable, controllable, dependable, and effective and that natural ventilation was difficult to control. In recent decades, studies of the theory and practice of natural ventilation have shown that natural ventilation can be all of these things—and very much cheaper. We have progressed to the point, in many applications, at which natural ventilation has been accepted as preferable to mechanical ventilation. For example, most new pig buildings in the United Kingdom have automatically controlled natural ventilation (ACNV). In the past, the reputation of natural ventilation suffered due to ignorance of theory and consequent inconsiderate design. Far too often, natural ventilation has been blamed for the failures of designers.

Natural ventilation systems rely on pressure differences to move fresh air through buildings. Pressure differences can be caused by wind or the buoyancy effect created by temperature differences or differences in humidity. In either case, the amount of ventilation will depend critically on the size and placement of openings in the building. It is useful to think of a natural ventilation system as a circuit, with equal consideration given to supply and exhaust. Openings between rooms such as windows, louvers, grills, or open plans are techniques to complete the airflow circuit through a building.

Wind can blow air through openings in the wall on the windward side of the building, and suck air out of openings on the leeward side and the roof (Figure 16). Temperature differences between warm air inside and cool air outside can cause the air in the room to rise and exit at the ceiling or ridge, and enter via lower openings in the wall. Similarly, buoyancy caused by differences in humidity can allow a pressurized column of dense, evaporative cooled air to supply a space, and lighter, warmer, humid air to exhaust near the top. These three types of natural ventilation effects are further described below.

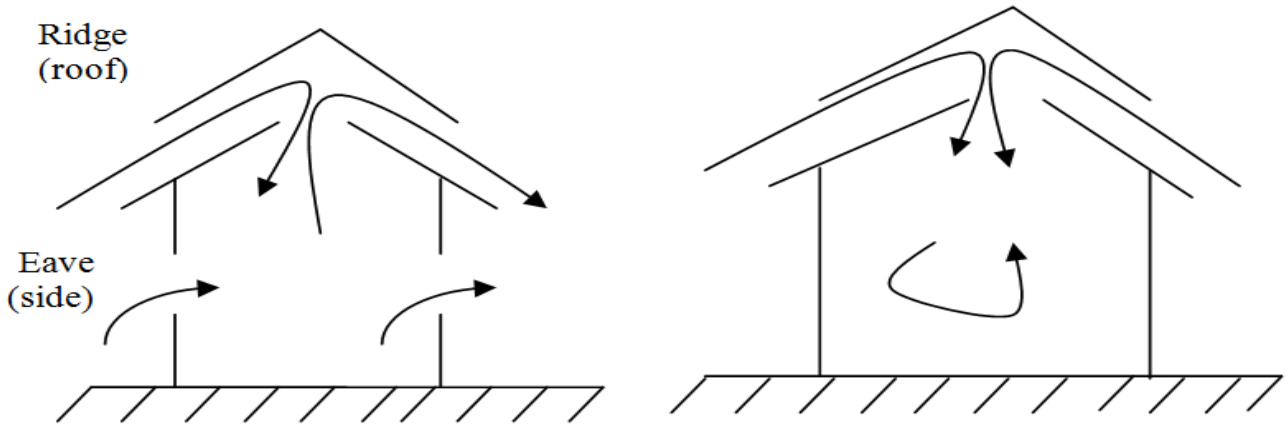


Figure 16: Roof and Side Ventilation

2-3.1.1 Natural ventilation by wind

Wind causes a positive pressure on the windward side and a negative pressure on the leeward side of buildings (Figure 17). To equalize pressure, fresh air will enter any windward opening and be exhausted from any leeward opening. In summer, wind is used to supply as much fresh air as possible while in winter, ventilation is normally reduced to levels sufficient to remove excess moisture and pollutants. An expression for the volume of airflow induced by wind is:

$$Q_{wind} = KAV \qquad 3.1$$

where
Q_{wind} = volume of airflow (m^3/h)
A = area of smaller opening (m^2)
V = outdoor wind speed (m/h)
K = coefficient of effectiveness

The coefficient of effectiveness depends on the angle of the wind and the relative size of entry and exit openings. It ranges from about 0.4 for wind hitting an opening at a 45° angle of incidence to 0.8 for wind hitting directly at a 90° angle. Sometimes wind flow prevails parallel to a building wall rather than perpendicular to it. In this case it is still possible to induce wind ventilation by architectural features or by the way a casement window opens. For example, if the wind blows from east to west along a north-facing wall, the first window (which opens out) would have hinges on the left-hand side to act as a scoop and direct wind into the room. The second window would hinge on the right-hand side so the opening is down-wind from the open glass pane and the negative pressure draws air out of the room. It is important to avoid obstructions between the windward inlets and leeward exhaust openings.

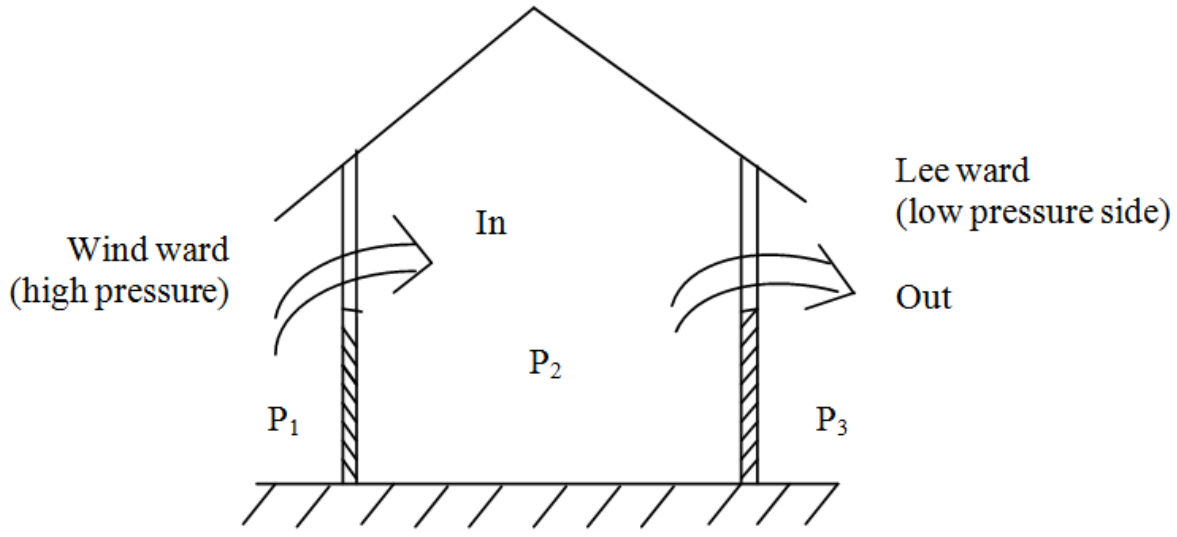


Figure 17: Ventilation by Wind

2-3.1.2 Natural ventilation by buoyancy effect

Buoyancy ventilation may be temperature-induced (stack ventilation) or humidity induced (cool tower). The two can be combined by having a cool tower deliver evaporatively cooled air low in a space, and then rely on the increased buoyancy of the humid air as it warms to exhaust air from the space through a stack. Buoyancy results from the difference in air density. The density of air depends on temperature and humidity (cool air is heavier than warm air at the same humidity and dry air is heavier than humid air at the same temperature). Within the cool tower itself the effect of temperature and humidity are pulling in opposite directions (temperature down, humidity up). Within the room, heat and humidity given off by occupants and other internal sources both tend to make air rise. The stale, heated air escapes from openings in the ceiling or roof and permits fresh air to enter lower openings to replace it. Stack effect ventilation is an especially effective strategy in winter, when indoor/outdoor temperature difference is at a maximum. Stack effect ventilation will not work in summer (wind or humidity drivers would be preferred) because it requires that the indoors be warmer than outdoors, an undesirable situation in summer. A chimney heated by solar energy can be used to drive the stack effect without increasing room temperature, and solar chimneys are very widely used to ventilate composting toilets in parks.

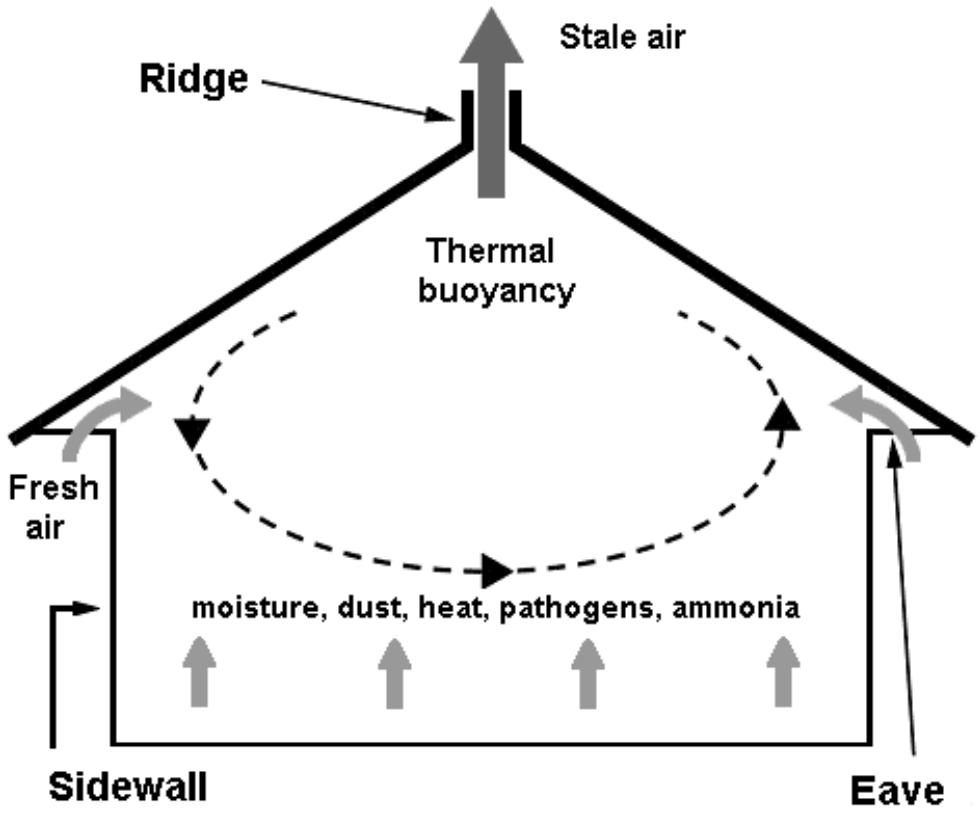


Figure 18: Ventilation by buoyancy effect

The rate of air movement resulting from this stack effect (buoyancy effect is directly proportional to the pressure difference and the difference in height between the inlet and outlet. The pressure difference may be converted to a temperature difference by the use of perfect gas laws:

$$V = \theta \left\{2gH\left(\frac{T_1 - T_0}{T_1}\right)\right\}^{1/2} \qquad 3.2$$

Where: V = discharge velocity (m/s)

θ = reducing factor/ constant (0.5 – 0.7)

g = acceleration due to gravity 9.81m/s^2

h = height difference between inlet and outlet (m)

T_0/T_1 = inside temperature and outside temperature

The reducing factor varies with structure, shape and also accounts for losses due to friction of the air against the inlet surface

Combining equation 3.1 and 3.2

$$\left(\frac{Q}{A}\right)^2 = \theta^2 2gH\left(\frac{T_1 - T_0}{T_1}\right)$$

2-3.2 Mechanical Ventilation

Mechanical ventilation is also referred to as forced ventilation. Forced ventilation means that the air-exchange rate in ventilated rooms is established by means of fans. It differs from natural ventilation in being less dependent on wind velocity and thermal buoyancy and in energy consumption.

Advantages

1. A more positive action is ensured.
2. Ventilation is less affected by wind
3. Ventilation can be easily controlled.

Disadvantages

1. Initial investment cost is high
2. High operational cost

Principles

A ventilation system always includes two main parts: inlet and outlet systems. The airflow pattern at an inlet is completely different from the airflow pattern at an outlet, because the air stream from an inlet is unidirectional and therefore able to travel a long distance before it is discharged, contrary to the conditions at the outlet, where the air is sucked in from all directions. Normally, air inlets are shaped as inlet valves or nozzles with well-defined jets.

In the case of forced ventilation at least one of the air inlets contains fans. If the fan is placed in the outlet system, it is called a *negative-pressure system*. If the fan is placed in the inlet, it is called a *positive-pressure system*, and if there are fans in both inlet and outlet it is called an *equal-pressure system*.

Exhaust or negative pressure system

The term *negative pressured ventilation* covers many different combinations of inlet and exhaust types. The inlets can e.g. be inlet valves in walls or ceilings, inlet slots along the overhang, distribution channels with nozzles, porous ducts or ceilings, and so forth. The exhaust fans can be placed in the outer wall, in the roof, or as a part of an exhaust system under a slatted floor. Common to all combinations is that fresh air is supplied to the room by a negative pressure created by exhaust fans. The size of the negative pressure depends on the used inlet velocity and the inlet air resistance. The advantages of negative pressured ventilation are the guarantee that humid indoor air will not pass walls and roof leakages and the fact that the energy consumption is low compared with equal pressured systems. The disadvantage is that the outdoor wind speed can influence the inlet jets, for example if wall inlets are used.

In summary, for negative pressure systems:

- Air is forced out of house through outlets
- Partial vacuum is then created inside the house and this
- Results in air pressure drift (gradient) through inside and outside the house (i.e. static pressure) which forces outside air into the house.

Equal Pressure system

Equal pressured ventilation means a system in which the capacities of the inlet and the exhaust fans at any time are the same. The inlet part normally consists of inlet ducts or heads with nozzles. It can eventually be combined with simple gable fans without any real distribution system. The exhaust fans can be of the same types as for a negative pressured system. The advantages are that the system is only slightly dependent on wind, and that the inlet velocity can be kept at a certain level regardless of the building tightness. Because the ventilation flow rate is well controlled, no unnecessary energy loss of artificial heat occurs. The disadvantages of the

system are that it is more complicated and costly to maintain, and that the energy consumption is approximately twice as high as for negative pressured systems.

Positive Pressure system

In *Positive pressured ventilation* systems the ventilation flow rate is controlled by fans and the outlet air leaves the chimneys by means of low positive pressures of a few pascals. The same types of inlet systems as those used for equal pressured ventilation can be used. The advantages are that the inlets are well controlled, as seen for equal pressured systems, and that the energy consumption is low. The disadvantage is the risk of pressing moist air into the building construction.

In summary, for positive pressure systems:

- Outside air is forced into the house through inlets
- Inside air pressure is increased
- Inside – outside air gradient forces air out of the house.

Exchanges in Ventilation process

(1) **Air exchange**

- O_2 (in)
- Harmful gases and undesirable odours (out)
- Air borne disease organisms

(2) **Moisture exchange**

-Removal or addition of moisture

(3**) Heat exchange**

Sensible heat removal or addition

Systems Components of mechanical ventilation

There are three main components

- Fans
- Air distribution system
- Controls to regulate fans

2-3.1.1 Fans

The fan is the "heart" of a mechanical ventilation system. It provides the driving force for the needed air exchange in livestock facilities. Without adequate air exchange, animal performance can suffer while moisture and manure gases will corrode metal building components and

equipment and sensing elements on thermostats, electrical switches and controls. Livestock building environments can be better managed by understanding the factors influencing ventilation fan performance. Required ventilation rates vary with number and size of animals, stocking density and inside and outside temperatures. During cold weather, minimum air changes are needed to remove moisture and harmful gases while conserving heat. As outside temperatures increases, additional airflow is needed to remove excessive heat.

Ventilation fans are used to develop pressure drifts below the inside and outside of an animal house. Fans installed in animal houses are subjected to various kinds of abuses:

- Exposure to dust particles.
- Exposure to corrosion.

Factors to consider when selecting fans include:

1. Fan type
2. Required ventilation capacity
3. Quantity of air delivered at different static pressures
4. System static pressure
5. Energy efficiency
6. Service and support
7. Durability, reliability, and expected life
8. Suitability for intended use
9. Cost

Fan types

Fans can be axial flow or centrifugal. Axial fans are most commonly used in livestock buildings. Axial flow fans pass air straight through the fan parallel to the fan blade drive shaft. Centrifugal fans (squirrel cage fans) bring air in through a centre inlet, the air makes a right angle turn and is discharged perpendicular to the fan axis. Axial flow fans are designed to operate at low static pressures. They are commonly used for livestock building ventilation. The initial cost is less and performance is influenced less by dirt build up on the fan blades. Propeller fans, tube axial fans and vane-axial fans are examples of types of axial fans.

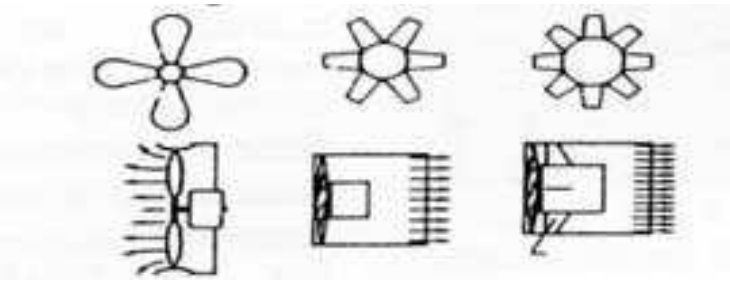


Figure 19: Axial fan types (propeller, tube-axialand vane-axial)

Propeller fans have propeller-shaped blades mounted in a circular ring or orifice plate and a drive motor. Propeller fans scoop air and move it away from the fan blade. Blade tip clearance is an important factor in fan performance. A small, uniform clearance is preferred to prevent air from flowing back around the propeller. These fans move large volumes of air at low static pressure. Propeller fans are typically 6” to 72” in diameter. Another type of propeller fan is the circulation fan used to help improve air distribution, reduce temperature stratification, increase air velocity in the animal zone, and eliminate dead air spots. Circulation fans usually operate at zero to low static pressure. Paddle fans mounted to the ceiling are an effective method to circulate large volumes of air against zero or low static pressure.

Tube-axial fans consist of a tube-shaped housing, a propeller-shaped blade, and a drive motor. Tube-axial fans can operate at higher static pressure because of a larger hub and reduced blade tip clearance, so less air flows back through the fan. Tube-axial fans are typically used in low and medium pressure duct air distribution systems. A rapidly growing use for this fan is manure pit ventilation. They are typically 12” to 48” in diameter.

Vane-axial fans similar in design and application to tube-axial fans. The major difference is that air straightening vanes are added either in front of or behind the blades to reduce the circular motion of the air. This results in a more efficient fan capable of operating at a slightly higher static pressure. Typical uses are grain drying, aeration, and sometimes pressure duct air distribution.

Centrifugal fans (Figure 20) are quieter and can operate at much higher static pressures than axial flow fans. Centrifugal fans operate by throwing air away from the fan blade tips. These blades can be forward curved, straight, or backward curved. Centrifugal fans should be used for handling relatively clean air because dirt accumulation on centrifugal fan blades affects efficiency and airflow capacity. Centrifugal fans are commonly used in grain drying and forced air heating systems where high static pressures are required.

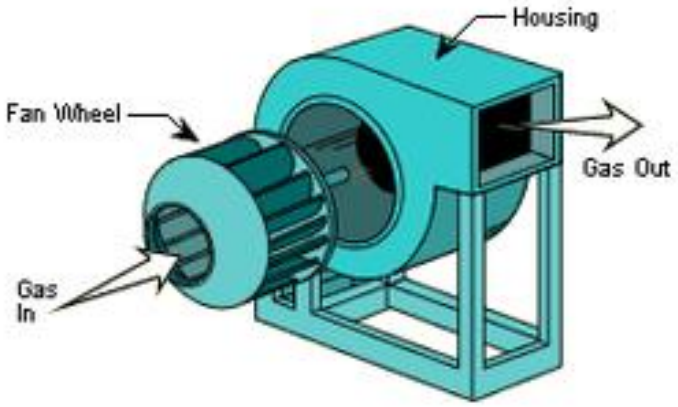


Figure 20: Centrifugal fan

Centrifugal fans are categorised into three according to the form of the blade (Figure 21). These are:

1. The forward curved blade fans
2. Radial blade
3. Backward curved

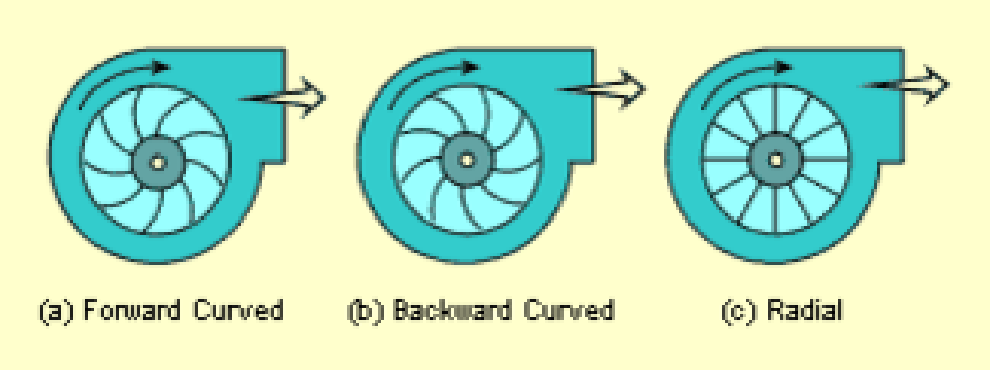


Figure 21: Categories of centrifugal fans

Ventilation Capacity

A ventilating system is used to control the environment. Ventilation capacity is based primarily on the amount of air exchange needed to remove moisture in the winter and excess heat in the summer. Airflow requirements vary with animal size and outside weather conditions. Ideally, ventilation air must vary from just enough air to maintain good air quality during very cold weather up to a maximum rate to eliminate heat stress during hot weather.

A properly sized ventilation system should provide at least three seasonal ventilation rates – cold, mild, and hot weather. Cold weather ventilation is the minimum continuous rate that provides enough air to remove excess moisture. Cold weather fans run continuously with an emergency shut-off thermostat. Mild weather ventilation increases the air exchange rate to modify temperature and remove moisture. Mild weather fans are typically turned on and off with thermostats to maintain a desired air temperature. Hot weather ventilation reduces heat build-up and increases air movement past the animals.

Static Pressure

Static pressure is the pressure that a fan is capable of developing. Static pressure is also a measure of the resistance to airflow or the pressure developed as a given rate of airflow moves through system. Static pressure or air pressure difference is usually measured between the inside and outside of a building with a manometer. Resistance to airflow occurs throughout the entire ventilation system. In a mechanically ventilated barn, the exterior air inlets, baffled slot inlets, building space, fan safety guard, fan housing, louvered shutters, and exterior weather hood all contribute to the system's resistance to airflow. For any ventilation system, the resistance to airflow or static pressure increases at higher airflow rates.

Fan Efficiency

Fan efficiency is a measure of volume of air moved per minute per watt of electrical power input. Energy efficient fans are usually more expensive. However, energy efficient fans can save the extra cost of the fan in about 2 to 3 years. Good quality fans can last more than ten years, the money saved is more than the cost of the fan.

If fan efficiency ratings are not available, the following guidelines can be helpful.

1. For a given airflow and static pressure, a large diameter fan is more energy efficient than a small diameter fan.
2. For a given ventilating rate, one large diameter fan is more efficient than several smaller diameter fans.
3. When two fans have the same diameter, the fan with the lower motor current input rating is usually more efficient.
4. If two fans deliver the same airflow at the same static pressure, the fan with the slower speed is usually quieter and more efficient.

Fan Motors and Drives

Fans can be direct or belt driven. There are several types of fan motors:

- Single speed type
- Double speed

- variable speed

Fan motors should be:

- Totally enclosed for dust protection
- Able to operate continuously at all times
- equipped with overload protection

Fan Laws (Similarity laws)

1. $Q/ND^3 = const\tan t$ which implies that $N_1/Q_1 = N_2/Q_2$
2. $P_w/N^3D^5P = const\tan t$ which implies that $Pw_1/N^3{}_1 = Pw_2/N^3{}_2$
3. $G/N^2D^2 = const\tan t$
4. $P_r/N^2D^2 = const\tan t$ which implies that $\mathrm{Pr}_1/N^2{}_1 = \mathrm{Pr}_2/N^2{}_2$

Fan efficiency, $Eff_{fan} = \dfrac{air\,power}{shaft\,power}$

Fan Power, $P_w = 9.81QP_{rt}K_p$

Q = Volume flow rate
Pr = Fan total pressure
Pw = power consumed
N = Air total speed or impeller speed
D = impeller diameter
g = action due to gravity
Ω = fluid density
Kp = component coefficient
P = density of air

Fan Ratings

Fans are rated and their ratings controlled by established institutions.

Speed (rpm)	**Diameter (cm)**	**Motor sys (Hp)**	**Fan capacity (m^3/min)**
1725	35.6	1/6	40
1725	45.7	1/3	95
906	76.0	½	195
695	91.0	½	256

Fans in various capacities and numbers may be combined to meet the ventilation needs.

Air inlet system

Area of opening for air flow can be determined by the use of the equation; Q = VA

V = Velocity

A = area of opening

Controls to regulate Fans

- Fan speed type
    - single speed unit
    - double speed unit
    - variable speed unit
- Thermostat control
    - Timer type
    - Combination of thermostat and timer type



Self Assessment 2-3

1. By aid of a diagram, briefly discuss ventilation by wind.
2. Discuss ventilation by buoyancy effect.
3. Discuss the different types of fans and their advantages.

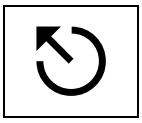


Answer tips

1. Read session 2-3.2.1
2. Read session 2-3.2.2
3. Read session 2-3.3.1



Learning Track Activities



Unit Summary

1. In this unit, we have defined and discussed the two main types of ventilation systems, natural and mechanical.

2. We have discussed the different types of fans that could aid the mechanical ventilation process.



Key terms/ New Words in Unit

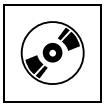
1. Ventilation
2. Buoyancy
3. Radiator
4. Positive pressure system
5. Axial fans
6. Centrifugal fans
7. Static pressure
8. Equal pressure system
9. Negative pressure system









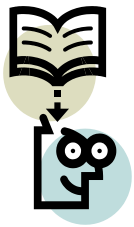


(choose an appropriate activity and give detail instructions)

Review Question:

1. A farm has 60 pigs housed in a 10m×40m×3m structure with $20m^2$ of windows and $12m^2$ of doors. It is 5℃ warmer inside and the heat produced per pig is 600W. U-values in $W/m^{2}℃$ are: window-5.8, door-1.5, ceiling-0.5 and wall-0.75. what percentage of the heat production should be removed by ventilation for heat balance to occur?

2. The walls of a pig house are constructed from a single 100mm solid block of thermal resistance $0.14m^{2}℃/W$ with 50mm polystyrene insulation of thermal resistance $1.25m^{2}℃/W$. find the overall heat coefficient.



Unit Assignments 3

1. Determine the air flow rate (ventilation rate) from a 30m long poultry building naturally ventilated that has a 0.008m eave inlet. Expected wind speed is 9m/s.
2. Air is removed through a structure at a rate of 2m3/s against a pressure of 275 Pa. Determine the required input Power of the fan that can do the job assuming 75% efficiency.
3. A fan is rotating at a speed of 15rev/s and moves 2m3 of air every second. If the speed is increased to 30rev/s,
   (a) What is the new amount of air moved per second?
   (b) If the initial pressure developed by the fan was 500 Pa, what is the new pressure developed?
4. A fan running at 500rev/min on a 6 KW motor produces 12m3/s of air at a pressure of 200 Pa. If the speed is doubled, determine
   i. the expected volume flow
   ii. Expected pressure and power
   iii. Will it be necessary to change the motor?
5. An air impeller have an efficiency of 82% when producing 10m3/s of air at 220 Pascal, 60% efficiency when producing 6m3/s at 330 Pascal and then 70% efficiency when producing 12m3/s at 100 Pascal. Calculate and compare the power ratings.

Unit 4

LIVESTOCK HOUSING

Introduction

This unit covers waste handling, treatment and disposal in livestock housing. It also covers emissions and how they can be handled in livestock housing.



Learning Objectives

After reading this unit you should be able to:

1. To describe the two forms of animal waste and describe the methods of handling waste.
2. To describe both aerobic and anaerobic treatment and list and describe the method by which solid and liquid waste may be treated.
3. Discuss alternative methods of animal waste ultimate disposal.
4. Estimate the storage tank or pit size for a given animal operations for a given waste production data and storage interval.

Unit content

Session 1-4: livestock housing

- 1-4.1 Introduction
- 1-4.2 Waste handling
- 1-4.3 Waste treatment
- 1.4.4 Waste disposal

Session 2-4 Animal environment monitoring

- 2-4.1 Emissions
- 2-4.2 Assessing animal environment

SESSION 1-4: ANIMAL WASTE MANAGEMENT CONCEPT

1-4.1 Introduction

Functional & spatial requirements (feeding, watering, resting, waste)
In planning a farm house, adequate space must be allowed for each of the daily activities.

Animal waste management concept
Waste management is a critical concern in many parts of the world where the expanding livestock and poultry industry is concentrating.

Waste management is necessary to:

1. maintain good animal health through sanitary facilities
2. avoid pollution problems (pollution of air and water)
3. comply with local and state regulations
4. avoid extra production cost and undesirable working situation

There are 3 basic aspects of animal waste management:

1. waste handling
2. waste treatment
3. waste disposal

1-4.2 Waste Handling

Animal waste may exist in 2 basic forms

1. Solid waste
2. Liquid waste

For solid waste, an accumulation of the solid parts of the waste material along with some liquid but where the major physical behaviour of the waste is as a solid. Waste material with moisture content (mc) of 40-50% wb or less would be considered as solid waste. For liquid waste, the solid and liquid components combine with or without additional water should have moisture content greater than 50%

In solid waste handling it is essential to have solid floors that can be bedded or drained. Solid waste handling is essentially a mechanical removal process noting that the mechanical operation is by scrapers, haulers, spreaders etc.

1-4.3 Waste Treatment

Waste treatment could be physical, chemical or biological. The focus here is on biological waste treatment. There are 2 basic methods of biological treatment of waste.

1. the aerobic treatment
2. the anaerobic treatment

Aerobic treatment

Occurs when there is sufficient dissolved O_2 and available in the waste to allow aerobic bacteria to breakdown the organic matter in the waste.

Anaerobic treatment

It is accomplished by anaerobic bacteria that utilise the O_2 in the organic matter itself as the organic matter in decomposed. No external oxygen is supplied.

Solid waste treatment

The major method of solid waste treatment is composting. Composting is accomplished by piling the waste and turning it frequently to provide aeration for aerobic bacteria to decompose while maintaining a high enough temperature within the pile to destroy pathogenic organisations weed seeds. The volume of composted waste may be between 30 - 60% of the volume of the original waste.

1-4.4 Waste Disposal

The ultimate disposal of animal waste material should not be accomplished in a manner that will endanger the health and well-being of people, animals and plants.

Waste ultimate disposal methods include the following:

1. land disposal, -evaporation
2. incineration and-re-cycling

Land disposal

This is the traditional practice of dropping waste. This practise, however, presents problems when large quantities of waste are generated and ultimately distributed over a limited land area. Solid waste can mechanically be distributed over the land. Liquid waste can be distributed by the use of pumps and other equipment.

Evaporation

Evaporation of the liquid waste of animals is an ultimate disposal means that has the greatest potential in an area where evaporation exceeds rainfall.

Incineration

Done in areas where there are no alternatives. Solid waste is burnt to ashes. It pollutes the environment and is therefore not a very good option.

Re-cycling

Recycling the waste material in whole or in part back to the animal production system is another ultimate disposal alternative. Current attempts have been geared towards the recycling of waste as a medium for bedding and as a feed material rich in nutrients. Methods under studies range from

simple mining of raw materials with new feed to complex processing that yields useable productions including proteins, vitamins and minerals.

Waste storage capacity

Waste storage capacity takes into consideration the number of animals involved, the daily waste product rate (see Table 3) and the number of storage damp before waste removal. The number of storage days is referred to as the Retention time. It can be 180 days in cold climates and 90 days for warm climates

Table 3: Animal waste production

Animal	**Dairy**	**Beef**	**Poultry**	**Swine**	**Sheep**
weight(kg)	600	350	2.0	90	45
Waste Production (m^3/kg)	0.04	0.03	0.002	0.008	0.003

Structure: Dome - $\frac{2}{3} r^3 \pi$

Cylindrical - $r^2 h \pi$

Rectangular - Lbh

2.4.2 **Assessing Animal environment**

Parameters for assessment

- Air Temperature (dry bulb, thermographs, thermisters etc)
- surface Temperature (thermographs)
- air moisture (humidity) (hygrometers, thermohydrographs)
- air movement (hot wire, rotating vane, cap-type anemometer)
- air pollutant
- Aerial gases (colorimetric indicator tube, spectroscopic types
- aerial dust (volume sampler method)
- solar and thermal radiation
- light intensity (photovoltaic meters
- sound levels
- air pressure



Self Assessment 1-4

1. Why is it necessary to manage waste in livestock housing?

2. Discuss the different forms of waste disposal.



Answer tips

1. Read introduction to session 1-4.1
2. Read session 1-4.4

SESSION 2-4: ANIMAL ENVIRONMENT MONITORING

2-4.1 Emissions – Pollutants

Air pollutants include solid, liquid and gases floating substances like dust /root, smoke, fog, unit, vapours, gases, heat.

Animal houses

In animal houses, sources of pollutants are

a) Dust
b) gases (NH_3, H_2, CO_2, NH_4) .heat
c) vapours (moisture)

Effects of Bad working climate

1. Reduction of comfort and ability to work (both workers and animals)
2. Reduction of productivity
3. Regular illness
4. Dizziness
5. Irritations of skin and eyes
6. Reduction of attentiveness

7. high accident rate

Table 4: Emission Limits in Livestock Housing

Agent	Limit Value (animal)	Limit value (human)
Ammonia (ppm)	10-25	25
CO_2 (ppm)	3000	5000
H_2S (ppm)	0.5	10
Dust (ppm)	10	5

ppm = parts per million

Ammonia

It is a tonic air pollutant most frequently uncounted in high concentration in animal houses. It has a characteristic pungent odour that humans can detect at 10ppm. Being less desire than air ($0.77g/am^3$ against $1.29g/cm^3$) it moves gradually throughout an enclosure. Ammonia is primarily an irritant and has been known to create health problems in animals in confinement buildings. Irritation of the eyes and respiratory tract are common problems from prolonged exposure to this gas. It is also associated with soil-acidification processes.

Sources of ammonia emissions

Major sources

- Storage
- Disposal
- Spread of animal waste

Minor sources

- Application of artificial fertilizer
- Others (refuse damps etc)

Production of ammonia

Ammonia is produced during substance processes like

- Feeding
- watering
- waste management and
- ventilation

$$CO(NH_2)_2 + H_2O \xrightarrow{\text{enzyme}} CO_2 + 2NH_3$$

Release and emission

The release of ammonia depends on the following:

- Temperature

- Wind speed
- surface area
- PH value

$NH_3 + H_2O \rightleftharpoons NH_4 + OH$

Factors that affect the production of ammonia

- Type of animal
- Animal activity
- Production system
- Management of waste during storage
- Duration of waste storage
- Degree of contaminants of floors
- Degree of contaminants of animals
- Type of feed

Health and environmental effects of ammonia

Farm workers

- Coughing
- Mucous secretion in the throat (nose running)
- Changes in health and heart beat

Animals

- Reduction in daily weight gain
- Loss of appetite
- Inability to convert feed effectively
- Cannibalism
- Irritation of skin and eyes
- Tears and fluids out of the eyes of animals
- High mortality rate.

Plants, soil, structure

- Ammonia reduces plant growth by affecting leaf formation.
- Ammonia causes soil acidity
- Ammonia is abrasive on structures

Reduction of ammonia production

Steps that can be used to reduce ammonia production could include:

- Keeping floors free from urine and faeces
- Keeping floors dry
- Reducing the surface area of the excreta
- Using optimal protein levels in animal feed

- Use of additives like acids to neutralise animal faeces

Hydrogen sulphide (H_2S)

Usual concentration of in animal houses should be less than 10ppm which is not toxic. H_2S is denser than air and has an odour of rotten egg. High concentration can cause the following:

- Irritating of respiratory tracts
- Coma and
- in severe cases Death

Dust

Combined organic and inorganic substances which can be suspended in the air. Dust normally range from feed particles less than 1µm to solid particles around 100µm. Acid dust inside animal houses varies with species, e.g. in poultry houses the dust contains a considerable amount of feathers and skin parts or debris. It also contains particles of feed, litter as well as soil. In pig houses dust is mainly from feed and sometime dried excreta. Dust becomes a source of worry when the size range is within the lung depositable size.

Dust as disease carrier

- Dust borne gases
- Dust borne microbes

2-4.2 Assessing Animal environment

Parameters for assessment

- Air Temperature (dry bulb, thermographs, thermisters etc)
- surface Temperature (thermographs)
- air moisture (humidity) (hygrometers, thermohydrographs)
- air movement (hot wire, rotating vane, cap-type anemometer)
- air pollutant
- Aerial gases (colorimetric indicator tube, spectroscopic types
- aerial dust (volume sampler method)
- solar and thermal radiation
- light intensity (photovoltaic meters
- sound levels
- air pressure



Self Assessment 2-4

1. What are the effects of a bad working environment?

2. Identify the sources and factors that affect the production of major emissions in livestock housing.

Answer tips

1. Read session 2-4.1
2. Read session 2-4.1

Learning Track Activities

Unit Summary

1. In this unit, we have discussed the forms of animal waste and available options for managing these waste streams. We have learnt about aerobic and anaerobic waste treatment methods for managing both solid and liquid wastes.

2. We have discussed emissions from livestock housing units and how these could be managed to provide conducive living environment for both humans and livestock.

Key terms/ New Words in Unit

1. Waste handling
2. Waste treatment
3. Waste disposal
4. Aerobic
5. Anaerobic
6. Emissions

(choose an appropriate activity and give detail instructions)

Review Question:

1. What are pollutants?
2. Name two sources of pollutants in an animal's house.
3. List four (4) factors that affect the production of Ammonia in an animal house.
4. Mention the effects of air pollutants in livestock houses.
5. What are the aims of animal environmental monitoring?
6. What consideration would you make in designing a system for monitoring the environment of animals?
7. Why is the incineration of waste discouraged in most parts of the world?



Unit Assignments 4

1) A waste treatment structure in the form of a dome is to be constructed for waste produced by 15 dairy and 20 beef cattle. Waste production in cubic metres per day is estimated for dairy as 0.3 and beef as 0.2. Determine the dimensions of the structure and comment on your answer.
2) A waste composting plant in the form of a dome is to be constructed for waste generated by 20 dairy and 30 beef cattle. Waste production for an animal in m3 / day is 0.3 and 0.15 for dairy and beef cattle respectively. Water that is used to wash the waste into the structure is estimated as 20% of the total waste generated per day. What dimension of the structure will you recommend if the storage time is four months?

Selected Answers to Unit Assignments

[Supply selected answers to Unit Assignments]

Course Quiz/Exams

[Supply course quiz of this course here for the attention of the Institute's examinations officer]

RESEARCH/PROJECT AREAS AND RELATED TOPICS IN THIS COURSE

SOME CASE STUDIES TO THIS COURSE

[Supply some case studies in this course for your students]

MY PAGE

Name: _____ Learning Centre: _____

Contact: Tel. _____ Email: _____ Emergency Name/Phone: _____

Important numbers: Student number _____ Examination number _____

Program: _____ Year: _____ Course code/title: _____

Course objectives: _____

_____

Course dates/Semester No ( ): Starts_____ Ends _____

FFFS schedule/Dates: _____

Quiz dates: _____

Assignments hand in dates: _____

Revision dates: _____

Group discussion/work members (names and contacts): _____

_____

_____

End of course Self Evaluation:
I have completed all Units & interactive sessions□, mastered all learning objectives, completed all self Assessments, unit summary, key words and terms, discussion questions, review questions, reading activity, web activity, unit assignments, and submitted all CA scoring assignments, learner feedback on this course and submitted my comments and course focus contributory questions to facilitator for discussion.

Self-grading: self assessment questions score _____ % Unit Assignments scored _____ %

My course conclusion remarks: _____

_____

_____

_____ (may continue on reverse side)

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